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Filtering market signals: dynamic asset allocation with momentum and hidden mean reversion

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We study dynamic asset allocation when returns display short-run momentum yet revert to a hidden long-run mean. We extend the two-factor specification of [Kojien, R.S., Rodriguez, J.C. and Sbuelz, A., Momentum and mean reversion in strategic asset allocation. *Manage. Sci.*, 2009, **55**, 1199–1213.] into a partially observable linear-Gaussian economy. We estimate the unobservable drift of the return process with a Kalman–Bucy filter and exploit the separation principle to characterize the optimal portfolio and its value under partial information. The optimal weight splits into a myopic momentum bet, an intertemporal hedge, and an information-hedging component that scales with the filter’s conditional error variance. Closed-form expressions for the indifference value of information show that the premium for perfect drift observability rises with the noise-to-signal ratio. A simulation study enables us to interpret our theoretical results, and a real data application reveals that the partial-information strategy behaves more smoothly, yet still outperforms a naïve benchmark.

Keywords: Momentum; Mean reversion; Kalman–Bucy filtering; Partial information; Portfolio optimization

JEL Classifications: C02, C61, G11

1. Introduction

Financial markets display a well-documented pattern of return persistence at intermediate horizons, i.e. the so-called momentum effect, and mean-reversion at both very short and longer horizons. Classic studies by Fama and French (1988) and Poterba and Summers (1988) show mean reversion of stock-index returns over investment periods exceeding one year, while Jegadeesh and Titman (1993) find that *past winners* keep outperforming over the subsequent 3- to 12-month window, indicating a momentum premium. These opposite effects imply that investors seeking to exploit momentum must pay attention to the longer-term mean reversion effect, and any asset-allocation framework that ignores either force is inevitably too simplistic.

Kojien *et al.* (2009) provide a tractable continuous-time model in which the expected excess return is a combination of an observable exponentially weighted momentum factor and an Ornstein–Uhlenbeck mean-reversion component.

Their full-information solution reveals that the optimal share of stock is not monotone in the investment horizon because near-term momentum, medium-term reversal risk, and long-run mean reversion interact in complex ways. The model, like most of the subsequent literature (see Section 1.1 for literature review), assumes that both state variables are directly observable. However, in practice, the momentum component is reconstructed from past prices, but the mean-reversion factor is latent and only imperfectly proxied by dividend yields or macro signals. Treating this hidden driver as observable overstates an investor’s information set and exaggerates portfolio aggressiveness. Such a shortcoming was already recognized by empirical users of the Kojien *et al.* (2009) framework and by earlier studies considering ‘unobservable drift’, which did not include momentum.

The present paper closes this gap by considering a setting that includes momentum and an unobservable mean-reversion, resulting in a joint filtering–control problem. We consider a continuous-time asset price model in which the drift consists of (i) a momentum component that captures short-term trend persistency and (ii) a mean-reverting

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component that represents a long-run fundamental anchor. The key novelty is that the mean-reversion level is unobservable (a ‘hidden’ state), reflecting uncertainty about the true fundamental value or long-term expected return. In particular, we extend the Koijen *et al.* (2009)’s two-factor model to a partially observable setting, with linear Gaussian dynamics, so that the unobservable drift can be estimated by a Kalman–Bucy filter. We solve the optimization problem via dynamic programming, making use of the separation principle, which guarantees equivalence between the original control problem and an auxiliary control problem under the observation filtration (see, e.g. Wonham 1968, Fleming and Pardoux 1982). Building on this insight, we obtain closed-form expressions for (i) the value function and (ii) the optimal portfolio weights under power utility. The optimal allocation decomposes naturally into a myopic demand that exploits current momentum, an intertemporal hedging term against future shifts in investment opportunities, and an information hedging term that arises due to uncertainty in estimating the mean-reversion factor.

With closed-form solutions in hand, we can compute the *indifference value of information*, that is the cash amount an investor would pay ex-ante to observe the drift perfectly. We recall the concept of value of information in the two-factor setting. Moreover, in the analytically convenient one-factor special case (obtained when the momentum-decay rate equals the mean-reversion speed), we derive a fully explicit formula, which highlights how the value of information rises with the noise-to-signal ratio and the relative weight of the momentum component. Our study also highlights the role of risk penalization, implemented as a volatility-weighted adjustment to terminal wealth, in shaping optimal behavior (see Rogers 2013). The penalty prevents extreme position-taking and smooths the intertemporal hedging and information hedging components of the optimal policy.

We complement the theoretical analysis with (i) a simulation study which incorporates an estimation of all model parameters under partial information, based on a simulated-annealing refinement of linear-Gaussian system identification; (ii) a real data application. Our simulation study confirms that the filter accurately tracks the hidden drift, even when parameters are estimated rather than known. Moreover, using daily S&P 500 data from 1991–2020, we compare the optimal decisions of two types of investors: a sophisticated investor who acknowledges partial information and a naïve investor who approximates the mean reversion factor by the log-dividend yield pretending to have full information. We find that the optimal strategy of the sophisticated investor is smoother than that of the naïve investor, yet it still outperforms the corresponding naïve benchmark, underscoring the economic relevance of accurately incorporating information in the model.

Our study connects three strands of literature that have largely developed in isolation: (a) momentum–reversal asset-pricing models, (b) portfolio choice under unobservable drift and (c) filtering techniques for hidden factors in finance. By integrating an observable momentum signal with a latent mean-reversion process, we provide a continuous-time model that yields a tractable optimization problem, simultaneously accounting for momentum, mean reversion, and partial

information, and resulting in closed-form optimal portfolio strategies. The latter indicates that the information-hedging term scales linearly with the conditional error variance of the Kalman filter. When the filter precision deteriorates, for example, due to the large volatility (noise variance) in the observation process or a larger conditional variance of the filtering error, the information-hedging component of the investment strategy increases, pushing the total position away from the pure momentum bet.

If estimates of the unobservable factor process are really uncertain, the strategy might even bet against the trend (e.g. sell even though prices are rising). That sets the stage for a big loss, say a ‘momentum crash’, if the market suddenly snaps back in the other direction. This mechanism aligns with the empirical evidence presented in Daniel and Moskowitz (2016), who demonstrate that the worst momentum losses occur when prior losers rally quickly after stress periods. This corresponds to a setup in which our model predicts elevated filtering uncertainty and oversized, crash-prone positions.

This paper is structured as follows. The introduction is complemented by a review of the literature in Section 1.1. In Section 2 we provide the underlying model setting. In Section 3, we solve the utility maximization problem for an agent having complete information. In Section 4, we study the optimization problem under partial information. Section 5 gives a detailed discussion of theoretical results. In Section 6, we investigate the special one-factor setting which results from a specific parameter choice. In Sections 7 and 8, we elaborate on model estimation, run a simulation study, and perform real-world data analysis.

1.1. Literature review

Recent empirical studies suggest that return autocorrelation has a persistent, yet horizon-dependent, pattern identified as momentum. Momentum was first rigorously established by Jegadeesh and Titman (1993) and is now widely observed for example in the context of global equities, bonds, commodities, and currencies (see, e.g. Moskowitz *et al.* 2012, Asness *et al.* 2013).

Incorporated into asset-pricing models via the *up minus down* (UMD)[†] factor of Carhart (1997), momentum delivers a sizeable premium. Still, it is prone to episodic crashes: sharp rebounds after market stress, such as the 2009 post-crisis rally, inflict large losses on trend strategies, quantified, e.g. by Daniel and Moskowitz (2016).

Behavioral theories such as Barberis *et al.* (1998) and Hong and Stein (1999) attribute these cycles to an initial underreaction, which drives trends, followed by an overreaction that triggers a reversal. Risk-based explanations emphasize compensation for liquidity or tail risk. On the other hand, mean reversion forms the long-run counterpoint: extreme performance reverses over multi-year horizons (De Bondt and Thaler 1985, Fama and French 1988, Poterba and Summers 1988), while microstructure frictions create short-horizon contrarian profits on the scale of weeks (Jegadeesh 1990, Lehmann 1990). Value

[†] Monthly premium on previous winners minus losers.

strategies, exemplified by the *high minus low* (HML) factor (monthly premium of the book-to-market), effectively monetize this long-term correction. In the time-series framework, Moskowitz *et al.* (2012) provide evidence that each asset's own past returns contain predictive power: a positive excess return over the past year predicts positive returns next month, but this trend-following signal fades and eventually reverses beyond a 12-month horizon. Such a time-series momentum effect, followed by mean reversion, underscores why multi-horizon strategies can outperform single-horizon ones. Recent studies have attempted to exploit this by dynamically adjusting investment horizons. For example, the study of He *et al.* (2018) provides an optimal asset allocation rule that combines short-run momentum with long-run reversal, and reports significant out-of-sample gains over pure momentum or pure contrarian strategies. The empirical evidence is thus characterized by a sequence of short-term bounce, mid-term momentum, and long-term reversal.

On the theoretical side, Koijen *et al.* (2009) present a tractable continuous-time framework in which the drift of the asset price comprises an exponentially weighted momentum factor M_t and an Ornstein-Uhlenbeck mean-reversion factor μ_t . This two-factor specification reproduces the horizon-dependent predictability observed empirically, namely momentum, which dominates in the short run, and mean reversion, in the long run. Moreover, it generates a non-monotonic optimal equity share: short- and very-long-horizon investors hold more stock than medium-horizon investors, who fear an impending trend reversal. The model includes earlier single-factor cases (e.g. the pure mean-reversion setup of Campbell and Viceira 1999) and shows that broad-market momentum adds little to long-term hedging demand. Similarly, optimal momentum-only strategies have been studied by Li and Liu (2022), who solved a dynamic portfolio problem with a risky asset exhibiting a momentum-driven drift (with known parameters).

Subsequent studies corroborate the value of combining the two signals: heterogeneous-agent simulations (Chiarella *et al.* 2007, Dieci and Westerhoff 2010) produce endogenous momentum–reversal cycles, while empirical work blending trend and contrarian strategies (Balvers and Wu 2006, Serban 2010) and cross-asset evidence from Asness *et al.* (2013) report superior risk-adjusted returns. Modern asset-pricing research, therefore, increasingly adopts dual-factor, regime-switching, or agent-based structures that view momentum and mean reversion as complementary facets of return predictability.

Seminal continuous-time studies by Lakner (1998) and Brendle (2006) showed that when a stock's drift is unobservable, yet linear-Gaussian, the investor can apply stochastic filtering (in particular the Kalman–Bucy filter) to address the optimization under restricted information, by enlarging the set of state variables with the filtered drift and its conditional variance. Markov modulated models with hidden Markov regimes (Bäuerle and Rieder 2005), applications of hidden regimes in pairs and convergence trading (Lee and Papanicolaou 2016, Altay *et al.* 2018, 2021), and expert-opinion updates (Frey *et al.* 2012) established verification theorems, clarified the separation principle, and quantified the utility cost of drift uncertainty. On the same strand of literature,

Gabih *et al.* (2024a) consider a continuous-time model where the stock return has an unobservable Gaussian mean-reverting drift, and investors receive noisy expert opinions at discrete time points.

Building on these themes, recent studies have attacked the momentum–reversal problem under partial information. A prominent example is Chen and Lee (2023), who solve an optimal-trading problem in which the drift is a single latent Ornstein–Uhlenbeck process observed only through a price spread. Thanks to the linear–Gaussian structure of the observation process and the unobservable drift, they apply the Kalman–Bucy filter and show that, for log-, exponential- and power-utility investors, the optimal weight is a simple function of the spot price and an exponentially weighted moving average (EWMA) of past prices—the EWMA emerges as the sufficient statistic for the hidden drift. Their specification, however, contains only one latent factor whose dynamics can resemble either momentum or mean reversion, depending on parameter values. We should remark here that, by contrast, our framework retains the EWMA-type interpretation while separating the two forces: the short-run momentum component is observable (constructed directly from past returns), whereas the long-run mean-reversion component is truly latent and must be filtered. This dual-factor structure enables us to investigate how an investor simultaneously exploits trend signals and hedges against hidden fundamental shifts—something the single-factor model of Chen and Lee (2023) may not account for.

2. Financial market

We consider a finite time interval $[0, T]$. We work on the filtered probability space $(\Omega, \mathcal{G}, \mathbb{G}, \mathbb{P})$, where $\mathbb{G} = (\mathcal{G}_t)_{t \geq 0}$ is the global filtration that satisfies the usual conditions; all processes we consider here are assumed to be $\mathbb{G}$ -adapted.

We consider a financial market with continuous trading where a cash account with value $(B_t)_{t \in [0, T]}$ and a risky stock with price process $(S_t)_{t \in [0, T]}$ are the available trading instruments. We assume that the dynamics of the cash account are given by

$$dB_t = rB_t dt,$$

with $B_0 > 0$, where $r \in \mathbb{R}$ is the risk-free interest rate. The price dynamics of the stock are driven by two stochastic factors M and μ , which represent the *momentum* and *mean reversion*, respectively. As μ is supposed to capture the long-term mean reversion in stock returns, we assume that it is stationary and that it follows an Ornstein-Uhlenbeck process. Hence, the mean reversion factor is given by

$$d\mu_t = \kappa(\bar{\mu} - \mu_t)dt + \sigma_\mu^\top dW_t, \quad (1)$$

which means that μ is mean reverting around the mean reversion level $\bar{\mu} \in \mathbb{R}$ with mean reversion speed $\kappa > 0$, and driven by $(W_t)_{t \in [0, T]} = (W_t^1, W_t^2)_{t \in [0, T]}$, which is a two-dimensional Brownian motion under the filtration $\mathbb{G}$, with volatility $\sigma_\mu \in \mathbb{R}^2$. Moreover, we assume that μ_0 has a Gaussian distribution with mean $\hat{\mu}_0 \in \mathbb{R}$ and variance $R_0 > 0$ (the case $R_0 = 0$ corresponds to full knowledge at $t = 0$ of the rate of return).

The economic interpretation of $\bar{\mu}$ is that it represents the long-run fundamental growth of the asset, around which the stationary process μ eventually fluctuates. The parameter κ determines how fast μ stabilizes around this growth rate, associated with the intrinsic value of the asset. The dynamics of the momentum factor is given by

$$dM_t = ((\phi - \omega)M_t + (1 - \phi)\mu_t)dt + \sigma_S^\top dW_t, \quad (2)$$

with $M_0 = 0$ and $\omega \geq \phi$. We model the stock price dynamics as the solution of the equation

$$\frac{dS_t}{S_t} = (\phi M_t + (1 - \phi)\mu_t)dt + \sigma_S^\top dW_t, \quad (3)$$

with $S_0 \in \mathbb{R}^+$, $\phi \in [0, 1)$, $\sigma_S \in \mathbb{R}^2$. Hence, one can see that the drift of the stock price depends on the two components M and μ , with relative weighting ϕ .

Note that momentum satisfies the equality

$$M_t := \int_0^t e^{-\omega(t-u)} \frac{dS_u}{S_u}, \quad (4)$$

which is consistent with the setting introduced in related finance literature (see, e.g. Koijen *et al.* 2009). More specifically, M is an exponentially weighted sum of past returns, where the weighting parameter ω controls the decay rate. A higher value of ω assigns greater weight to more recent returns, while reducing the influence of older returns. Moreover, we also observe that if $\omega > \phi^\dagger$

$$dM_t = (\omega - \phi) \left(\frac{(1 - \phi)}{(\omega - \phi)} \mu_t - M_t \right) dt + \sigma_S^\top dW_t.$$

This means that conditional on μ_t , the process M follows an Ornstein-Uhlenbeck process with mean reversion level $\frac{(1-\phi)}{(\omega-\phi)}\mu_t$ and mean reversion speed $(\omega - \phi)$. This leads to the economic interpretation that the long-run ‘percentage drift’ of S is

$$\left(\frac{\phi(1 - \phi)}{(\omega - \phi)} + (1 - \phi) \right) \bar{\mu}.$$

Note that in the case where $\omega = 1$, the long-run growth-rate of S is given by $\bar{\mu}$, which is related to the fundamental value of the asset.

REMARK 2.1 (i) We deal with a two-factor model that accommodates two important features of stock price dynamics. Indeed, returns tend to show a persistent behavior inherited from the past, accounted by momentum, and a mean reversion around a certain target mean. It is important to stress that momentum is a relevant effect over shorter time horizons, whereas mean reversion is visible over longer time horizons. Due to these considerations, the target mean is typically non-constant and subject to changing market conditions. Moreover, it is hard to observe the underlying

variables that affect the level of returns. These observations motivate our model under partial information. We will elaborate more on this in Section 4.1.

We also mention that a special case of the current setting occurs if one works under the parameter restriction that $\omega = \kappa$. In that case, the two-factor model reduces to one-factor. This special setup will be discussed in detail in Section 6.

(ii) Note that the dynamics of the triple (μ, M, S) can be equivalently written as

$$\begin{aligned} d\mu_t &= \kappa(\bar{\mu} - \mu_t)dt + \|\sigma_\mu\| dU_t^\mu, \\ dM_t &= ((\phi - \omega)M_t + (1 - \phi)\mu_t)dt + \|\sigma_S\| dU_t^S, \\ \frac{dS_t}{S_t} &= (\phi M_t + (1 - \phi)\mu_t)dt + \|\sigma_S\| dU_t^S, \end{aligned}$$

where processes $U_t^\mu := \frac{\sigma_\mu^{(1)}}{\|\sigma_\mu\|} W_t^{(1)} + \frac{\sigma_\mu^{(2)}}{\|\sigma_\mu\|} W_t^{(2)}$ and $U_t^S := \frac{\sigma_S^{(1)}}{\|\sigma_S\|} W_t^{(1)} + \frac{\sigma_S^{(2)}}{\|\sigma_S\|} W_t^{(2)}$ are one dimensional Brownian motions with correlation $\rho = \frac{\sigma_\mu^\top \sigma_S}{\|\sigma_\mu\| \|\sigma_S\|}$.

We consider an investment strategy $(h_t)_{t \in [0, T]}$, where h_t is the fraction of wealth invested in the stock at time t , and $1 - h_t$ is the fraction of wealth invested in the cash account at time t . We allow for short-selling the risky asset and borrowing from the cash account, hence $h_t \in \mathbb{R}$.

For every $(h_t)_{t \in [0, T]}$ we let $X_t^{(h)}$ be the value of the investor’s portfolio at time t , whose dynamics are given by

$$\frac{dX_t^{(h)}}{X_t^{(h)}} = (r + h_t(\phi M_t + (1 - \phi)\mu_t - r))dt + h_t \sigma_S^\top dW_t,$$

and with $X_0^{(h)} = x_0 > 0$.

To discourage a portfolio manager (acting on behalf of the investor) from taking excessively risky positions, the investor agrees to compensate the manager at time T with an amount penalized according to the realized volatility of the portfolio. This approach follows the framework in Rogers (2013, Section 2.22). Accordingly, the investor agrees to compensate the manager at time T with the amount $Z^{(h)} = (Z_t^{(h)})_{t \in [0, T]}$, the risk penalized wealth, which satisfies

$$Z_T^{(h)} = X_T^{(h)} \exp \left(-\frac{1}{2} \varepsilon \int_0^T \|\sigma_S\|^2 h_u^2 du \right),$$

with the penalization parameter $\varepsilon \geq 0$. It is easily seen that the dynamics of $Z^{(h)}$ is given by

$$\begin{aligned} \frac{dZ_t^{(h)}}{Z_t^{(h)}} &= \left(r + h_t(\phi M_t + (1 - \phi)\mu_t - r) - \frac{1}{2} \varepsilon \|\sigma_S\|^2 h_t^2 \right) dt \\ &\quad + h_t \sigma_S^\top dW_t, \end{aligned}$$

and with $Z_0^{(h)} = x_0 > 0$.

The manager’s goal is to maximize the expected utility of risk-penalized terminal wealth. Assuming that the investor’s preferences are described by a power utility function with

[†] Note that, if $\omega = \phi$ the dynamics of M becomes $dM_t = (1 - \phi)\mu_t dt + \sigma_S^\top dW_t$, $M_0 = 0$ and does not have any mean-reverting property.

risk aversion parameter $\gamma < 1$, the manager's optimization problem is to maximize

$$\mathbb{E} \left[\frac{(Z_T^{(h)})^\gamma}{\gamma} \right],$$

over the set of admissible strategies.

REMARK 2.2 For simplicity, we assume that the investor does not charge an assets-under-management fee, though incorporating a constant fee would be straightforward. The purpose of the volatility penalty is to incentivize more conservative strategies. In fact, it can be interpreted as a form of quadratic transaction cost. An important consequence of tying compensation to $Z_T^{(h)}$ is that it effectively modifies the manager's risk aversion from γ to $\gamma - \varepsilon$, thereby aligning the manager's incentives with the investor's preference for reduced risk. In the special case where $\varepsilon = 0$, the problem reduces to the classical CRRA utility maximization problem without risk penalization.

In the sequel, we address the optimization problem introduced above under different assumptions on the available information in the market. First, we assume that investors can observe any source of randomness that drives the market. This means that they observe stock prices and the drift of the underlying price process. We refer to this as the *full information* setting. Second, we suppose that investors observe the stock prices only. Therefore, they have no direct information on the mean reversion process in the drift, which can only be estimated given the observations. This case is called the *partial information* setting. Our goal is to discuss the form of optimal trading strategies and to compare value functions in these two cases.

3. Utility maximization under full information

We start our analysis with optimization problem under full information. We assume that investors have privileged information, that is, they can directly observe the stock prices, therefore momentum, and the mean reversion process dynamics. The available information is represented by the global filtration $\mathbb{G}$, which is the natural filtration generated by the two-dimensional Brownian motion W , completed with $\mathbb{P}$ -null sets. Our results revisit and extend the findings of Koijen *et al.* (2009) within the context of the risk-penalization.

For the given setting, we first define the set of *admissible strategies* as follows.

DEFINITION 1 A $\mathbb{G}$ -admissible portfolio strategy is a self-financing and $\mathbb{G}$ -adapted strategy $(h_t)_{t \in [0, T]}$ that satisfies integrability condition

$$\mathbb{E} \left[\int_0^T h_t^2 dt \right] < \infty. \quad (5)$$

We denote by $\mathcal{A}^{\mathbb{G}}$ the set of all $\mathbb{G}$ -admissible strategies.

REMARK 3.1 Note that condition (5), together with the assumption that $\kappa \geq 0$ and $\omega \geq \phi$, and the fact that the pair

$(M_t, \mu_t)_{t \in [0, T]}$ is a two-dimensional affine process, implies that

$$\mathbb{E} \left[\int_0^T (h_t^2 + |h_t M_t| + |h_t \mu_t|) dt \right] < \infty.$$

holds. Hence, the wealth process is well-defined and integrable.

Let $\mathbb{E}^{t, z, \mu, m}$ indicate the conditional expectation given the initial conditions $Z_t = z$, $\mu_t = \mu$ and $M_t = m$. Then, the fully informed investor aims to solve

$$\max_{h \in \mathcal{A}^{\mathbb{G}}} \mathbb{E}^{t, z, \mu, m} \left[\frac{(Z_T^{(h)})^\gamma}{\gamma} \right]. \quad (6)$$

This problem is addressed via the dynamic programming approach. Let $V^{\text{full}}(t, z, \mu, m)$ be the value function, that is

$$V^{\text{full}}(t, z, m, \mu) = \max_{h \in \mathcal{A}^{\mathbb{G}}} \mathbb{E}^{t, z, \mu, m} \left[\frac{(Z_T^{(h)})^\gamma}{\gamma} \right].$$

For any given $h \in \mathcal{A}^{\mathbb{G}}$, the risk-penalized wealth $Z^{(h)}$ is a controlled process. However, for simplicity of the notation, from now on we suppress the dependence on h and write Z instead of $Z^{(h)}$.

We introduce the following notation: let

$$K = \begin{pmatrix} -(\omega - \phi) & 1 - \phi \\ 0 & -\kappa \end{pmatrix}, \quad \xi = \begin{pmatrix} \phi \\ 1 - \phi \end{pmatrix},$$

$$\kappa_0 = \begin{pmatrix} 0 \\ \kappa \bar{\mu} \end{pmatrix},$$

and we define the vector

$$\Sigma = \left(\|\sigma_S\|, \frac{\sigma_S^\top \sigma_\mu}{\|\sigma_S\|} \right)^\top.$$

The theorem below characterizes both the optimal investment strategy and the corresponding value function.

THEOREM 3.2 (Value function and optimal strategy under complete information) *Suppose that the functions $A : [0, T] \rightarrow \mathbb{R}^{2 \times 2}$, $B : [0, T] \rightarrow \mathbb{R}^{2 \times 1}$ and $C : [0, T] \rightarrow \mathbb{R}$ are the unique solution of the system of ordinary differential equations (A1)–(A2)–(A3) on $[0, T]$. Then, for an investor with CRRA utility preferences with risk aversion coefficient $\gamma < 1$, and risk penalization parameter $\varepsilon \geq 0$ the value function is of the form*

$$V^{\text{full}}(t, z, m, \mu) = \frac{z^\gamma}{\gamma} \exp \left(\frac{1}{2} (m, \mu) A(t) \begin{pmatrix} m \\ \mu \end{pmatrix} + B(t)^\top \begin{pmatrix} m \\ \mu \end{pmatrix} + C(t) \right).$$

Moreover, the optimal investment strategy is a stochastic process with Markovian feedback form $(h_t^*)_{t \in [0, T]} = (h^*(t, M_t, \mu_t))_{t \in [0, T]}$ where the function $h^*(t, m, \mu)$ is

$$h^*(t, m, \mu) = \frac{\|\sigma_S\| \Sigma^\top B(t) - r + \left(\xi + \frac{1}{2} (A(t) + A(t)^\top) \Sigma \|\sigma_S\| \right) \begin{pmatrix} m \\ \mu \end{pmatrix}}{(1 - (\gamma - \varepsilon)) \|\sigma_S\|^2}. \quad (7)$$

Proof The proof is provided in Appendix 1.

REMARK 3.3 (Admissibility of the optimal strategy) Under the assumptions of Theorem 3.2, the strategy $(h_t^*)_{t \in [0, T]} = (h^*(t, M_t, \mu_t))_{t \in [0, T]}$ where $h(t, m, \mu)$ is given in (7), is $\mathbb{G}$ -admissible in the sense of Definition 1.

Indeed, by construction, M and μ are $(\mathbb{G}_t)$ -adapted continuous processes, and $h^*(t, m, \mu)$ is a Borel function of (t, m, μ) . Hence $(h_t^*)_{t \in [0, T]}$ is $\mathbb{G}$ -adapted and self-financing by construction of the wealth dynamics, so it remains to check the integrability condition.

Since A and B solve the system of ODEs (A1)–(A2)–(A3) with continuous coefficients on the finite interval $[0, T]$, standard ODE theory implies that A and B are continuous and thus bounded on $[0, T]$ [†]. Consequently, we can rewrite h^* in the affine form

$$h^*(t, m, \mu) = a_0(t) + a_1(t)m + a_2(t)\mu,$$

where the scalar functions a_0, a_1, a_2 are continuous and therefore bounded on $[0, T]$. Thus, there exist constants $C_0, C_1 > 0$ such that $|a_0(t)| \leq C_0$ and $|a_1(t)| + |a_2(t)| \leq C_1$, for all $t \in [0, T]$. Using the elementary inequality $(x + y + z)^2 \leq 3(x^2 + y^2 + z^2)$, we obtain

$$|h^*(t, M_t, \mu_t)|^2 \leq 3C_0^2 + 3C_1^2(M_t^2 + \mu_t^2), \quad t \in [0, T].$$

By assumption, μ and M are Ornstein–Uhlenbeck process, hence, they have finite second moments and $\sup_{t \in [0, T]} \mathbb{E}[\mu_t^2] < \infty$, $\sup_{t \in [0, T]} \mathbb{E}[M_t^2] < \infty$. Therefore

$$\mathbb{E} \left[\int_0^T |h_t^*|^2 dt \right] \leq 3C_0^2 T + 3C_1^2 \int_0^T \mathbb{E}[M_t^2 + \mu_t^2] dt < \infty,$$

since the integrand is bounded on $[0, T]$. This implies that h^* is $\mathbb{G}$ -admissible.

REMARK 3.4 (Existence of a solution of ODEs (A1)–(A2)–(A3) in $[0, T]$) The existence of a solution to the system of Riccati equations (A1)–(A2)–(A3) on the interval $[0, T]$ in the one-dimensional case has been investigated in several works, starting from the seminal contribution of Kim and Omberg (1996) (see also Colaneri *et al.* 2021 for the partial-information setting). Identifying sufficient conditions in the presence of multiple factors, however, is more challenging, due to the complicated structure of the Riccati ODE system for the functions A, B, C . In this context, we refer to Wonham (1968, Theorem 4.1) (see also Gabih *et al.* 2024b). In particular, the main difficulty lies in establishing the existence of the solution of the Riccati equation for $A(t)$ on $[0, T]$, as functions B and C would satisfy (multidimensional) linear ODEs with continuous coefficients.

[†] In Theorem 3.2 we assume that the system of Riccati equations has a well behaved solution in $[0, T]$. Proving this assumption is pretty standard in the one dimensional case but much more complicated in the multidimensional case. For results on the multidimensional (multiasset) case we refer to Gabih *et al.* (2024b).

■ 4. Utility maximization under partial information

We now assume that investors observe the stock price process S , but not the drift process μ . This assumption reflects the fact that the long-run mean-reversion level of a stock is *de facto* unobservable. In the existing literature, as e.g. in Koijen *et al.* (2009), proxies are used to estimate the unobservable mean-reversion component. A widely adopted approach is to assume that the mean-reversion factor is an affine function of log-dividends.

In contrast, our approach explicitly models the latent nature of the mean-reversion process and analyzes the resulting portfolio choice problem under partial information.

The goal of the investor is to characterize investment strategies that maximize the expected utility from terminal penalized wealth. Naturally, investment decisions can depend only on the information available to the investor at time t . We assume that the available information is given by the natural filtration of S . We denote this filtration by $\mathbb{F}$. Formally, we have that $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$, where $\mathcal{F}_t = \sigma\{S_u, 0 \leq u \leq t\}$, and $\mathcal{F}_t \subset \mathcal{G}_t$, for all $t \in [0, T]$. We assume that the filtration $\mathbb{F}$ satisfies the usual hypotheses of completeness and right continuity. We also stress that the information carried by S and M is the same, i.e.

$$\mathcal{F}_t = \sigma\{S_u, 0 \leq u \leq t\} = \sigma\{M_u, 0 \leq u \leq t\}.$$

Accordingly, in what follows, we consider M as the observation process.

REMARK 4.1 To show that the natural filtration of M and the natural filtration of S carry the same information, we observe that, by definition (see equation (4)) M_t is $\mathcal{F}_t$ -measurable, which implies that $\sigma\{M_u, 0 \leq u \leq t\} \subseteq \mathcal{F}_t$. Moreover, the converse inclusion, i.e. $\mathcal{F}_t \subseteq \sigma\{M_u, 0 \leq u \leq t\}$ holds since $dS_t/S_t = dM_t - \omega M_t dt$.

In the next step, we discuss the filtering problem, which allows us to determine the conditional distribution of the unobservable factor process, given the observations.

4.1. Partial information and filtering

As the signal and observation processes form a two-dimensional Gaussian system (μ_t, M_t) , $0 \leq t \leq T$, the filtering equations will be deduced by using the idea of Kalman–Bucy filter.

Let us denote the optional projection of μ with respect to filtration $\mathbb{F}$ by $\hat{\mu}_t = \mathbb{E}[\mu_t | \mathcal{F}_t]$, a.s., for every $t \in [0, T]$. Moreover, we denote by $R(t) = \mathbb{E}[(\mu_t - \hat{\mu}_t)^2 | \mathcal{F}_t]$, the mean square error of estimation of filtering. The next theorem states the filter dynamics.

PROPOSITION 4.2 (Kalman–Bucy Filter) *Let (μ_t, M_t) , $0 \leq t \leq T$, be a two-dimensional process satisfying the system of equations in (1)–(2) with $\mu_0 \sim N(\bar{\mu}_0, R_0)$. Then the conditional mean $\hat{\mu}_t$ and the mean square error $R(t)$ satisfy*

$$d\hat{\mu}_t = \kappa(\bar{\mu} - \hat{\mu}_t)dt + \frac{\sigma_S^\top \sigma_\mu + (1 - \phi)R(t)}{\|\sigma_S\|} dI_t, \quad (8)$$

$$\begin{aligned} \frac{dR(t)}{dt} = & -\frac{(1-\phi)^2}{\|\sigma_S\|^2} R^2(t) - 2 \left(\kappa + \frac{\sigma_S^\top \sigma_\mu (1-\phi)}{\|\sigma_S\|^2} \right) R(t) \\ & + \frac{(\sigma_\mu^{(1)} \sigma_s^{(2)} - \sigma_\mu^{(2)} \sigma_s^{(1)})^2}{\|\sigma_S\|^2}, \end{aligned} \quad (9)$$

with initial conditions $\hat{\mu}_0$ and $R(0) = R_0$, where I (innovations process) given by

$$dI_t = \frac{dM_t - ((1-\phi)\hat{\mu}_t - (\omega - \phi)M_t)dt}{\|\sigma_S\|}$$

is an $(\mathbb{F}, \mathbb{P})$ -Brownian motion.

Proof Since the observation and signal processes form a Gaussian system, and we are in the framework of Kalman-Bucy filters. Therefore, we apply Theorem 10.3 given in Liptser and Shiryaev (2013) and obtain the claimed result. ■

REMARK 4.3 We want to make a couple of considerations on the solution of the Riccati equation for the conditional variance.

- (i) For $R_0 \geq 0$, the Riccati equation (9) has a unique non-negative solution (see, e.g. Filipović 2009 [Lemma 10.12]). If we let

$$\Theta = 2 \sqrt{\left[\left(\kappa + \frac{\sigma_S^\top \sigma_\mu (1-\phi)}{\|\sigma_S\|^2} \right)^2 + \frac{(1-\phi)^2}{\|\sigma_S\|^4} (\sigma_\mu^{(1)} \sigma_s^{(2)} - \sigma_\mu^{(2)} \sigma_s^{(1)})^2 \right]},$$

then we get

$$\begin{aligned} R(t) = & -\frac{-\frac{2(\sigma_\mu^{(1)} \sigma_s^{(2)} - \sigma_\mu^{(2)} \sigma_s^{(1)})^2}{\|\sigma_S\|^2} (e^{\Theta t} - 1)}{\Theta (e^{\Theta t} + 1) + 2 \left(\kappa + \frac{\sigma_S^\top \sigma_\mu (1-\phi)}{\|\sigma_S\|^2} \right) (e^{\Theta t} - 1)} \\ & + \frac{\frac{2(1-\phi)^2}{\|\sigma_S\|^2} (e^{\Theta t} - 1) R_0}{\Theta (e^{\Theta t} + 1) + 2 \left(\kappa + \frac{\sigma_S^\top \sigma_\mu (1-\phi)}{\|\sigma_S\|^2} \right) (e^{\Theta t} - 1)} \\ & + \frac{\left(\Theta (e^{\Theta t} + 1) - 2 \left(\kappa + \frac{\sigma_S^\top \sigma_\mu (1-\phi)}{\|\sigma_S\|^2} \right) (e^{\Theta t} - 1) \right) R_0}{\Theta (e^{\Theta t} + 1) + 2 \left(\kappa + \frac{\sigma_S^\top \sigma_\mu (1-\phi)}{\|\sigma_S\|^2} \right) (e^{\Theta t} - 1)} \\ & + \frac{\frac{2(1-\phi)^2}{\|\sigma_S\|^2} (e^{\Theta t} - 1) R_0}{\Theta (e^{\Theta t} + 1) + 2 \left(\kappa + \frac{\sigma_S^\top \sigma_\mu (1-\phi)}{\|\sigma_S\|^2} \right) (e^{\Theta t} - 1)}. \end{aligned}$$

- (ii) If $R(0) = \bar{R}$ with $\bar{R} = \frac{\|\sigma_S\|^2}{(1-\phi)^2} \left(\kappa + \frac{\sigma_S^\top \sigma_\mu (1-\phi)}{\|\sigma_S\|^2} + \frac{\Theta}{2} \right)$, then $\frac{dR(t)}{dt} = 0$ and $R(t) = \bar{R}$ for all $t \in [0, T]$. This corresponds to taking the initial conditional variance equal to the steady state solution of the equation (9).
- (iii) If $R(0) > \bar{R}$, then it holds that $R(t)$ is non-increasing and $R(t) \in [\bar{R}, R(0)]$, for all $t \in [0, T]$. If $R(0) < \bar{R}$, then it holds that $R(t)$ is non-decreasing and $R(t) \in [R(0), \bar{R}]$, for all $t \in [0, T]$. That is, $R(t)$ is bounded from below and bounded above.

4.2. Optimization under partial information

The optimization problem under partial information involves a new ingredient: the investor can not directly observe the instantaneous drift of the asset, but rather only S , so that the strategy h can only be $\mathbb{F}$ -adapted, which leads to a different definition of *admissible strategies*.

DEFINITION 2 An $\mathbb{F}$ -admissible portfolio strategy is a self-financing and $\mathbb{F}$ -predictable strategy h that satisfies the integrability condition (5). $\mathcal{A}^\mathbb{F}$ is the set of all $\mathbb{F}$ -admissible strategies.

The results of Section 3 do not allow us to account for the partial information setting. The idea for solving the optimization problem under partial information is to formulate it in terms of observation-dependent quantities only, and apply the dynamic programming principle for the resulting *completely observable* control problem. This procedure is often referred to as the *separation principle* (see, e.g. Wonham 1968, Fleming and Pardoux 1982, Gennotte 1986†), and a strategy which is a function of the filter is called a *separated control*.

To begin, we express the momentum process M , and the penalized wealth process $Z^{(h)}$, for any strategy $h \in \mathcal{A}^\mathbb{F}$, with respect to observation filtration $\mathbb{F}$:

$$dM_t = ((1-\phi)\hat{\mu}_t - (\omega - \phi)M_t) dt + \|\sigma_S\| dI_t, \quad (10)$$

$$\begin{aligned} \frac{dZ_t^{(h)}}{Z_t^{(h)}} = & \left(r + h_t (\phi M_t + (1-\phi)\hat{\mu}_t - r) - \frac{\varepsilon}{2} h_t^2 \|\sigma_S\|^2 \right) dt \\ & + h_t \|\sigma_S\| dI_t. \end{aligned} \quad (11)$$

Both processes are driven by the innovation process I , which is a Brownian motion under the observation filtration $\mathbb{F}$. Hence, we can consider the triple $(Z^{(h)}, \hat{\mu}, M)$ of $\mathbb{F}$ -adapted processes as the state processes and formulate the optimization problem

$$\max_{h \in \mathcal{A}^\mathbb{F}} \mathbb{E}^{t, z, \hat{\mu}, m} \left[\frac{(Z_T^{(h)})^\gamma}{\gamma} \right], \quad (12)$$

where $\mathbb{E}^{t, z, \hat{\mu}, m}$ denotes the conditional expectation given initial conditions $Z_t^{(h)} = z$, $\hat{\mu}_t = \hat{\mu}$, and $M_t = m$ and $\gamma < 1$.

We define the value function corresponding to problem (12) as

$$V(t, z, m, \hat{\mu}) := \sup_{h \in \mathcal{A}^\mathbb{F}} \mathbb{E}^{t, z, \hat{\mu}, m} \left[\frac{(Z_T^{(h)})^\gamma}{\gamma} \right].$$

For the simplicity of notation, in the sequel we suppress the dependence on h and write Z instead of $Z^{(h)}$. We address the optimal portfolio problem using dynamic programming. We

† Wonham (1968) showed that in the case of quadratic cost and a Gaussian linear system with uncorrelated noise, the solution of (12) is again given by the solution of the HJB equation associated to (6), where in the definition of the Hamiltonian μ is replaced by $\hat{\mu}$. Of course, this is not the setting we are dealing with, as our cost depends on the CRRA utility, and we have that the driving noise of S and μ may be correlated. In a later paper (Gennotte 1986) studied exactly this set-up for general utility functions and the correlated, conditionally Gaussian case, which justifies the use of the *separation principle* in our work as well.

recall the definitions:

$$K = \begin{pmatrix} -(\omega - \phi) & 1 - \phi \\ 0 & -\kappa \end{pmatrix}, \quad \xi = \begin{pmatrix} \phi \\ 1 - \phi \end{pmatrix},$$

$$\kappa_0 = \begin{pmatrix} 0 \\ \kappa \bar{\mu} \end{pmatrix},$$

and we denote the vector

$$\widehat{\Sigma}(t) = \left(\|\sigma_S\|, \frac{\sigma_S^\top \sigma_\mu + (1 - \phi)R(t)}{\|\sigma_S\|} \right)^\top.$$

The next theorem provides the solution to the optimal control problem.

THEOREM 4.4 (Value function and optimal strategy under partial information) *Suppose that the functions $\widehat{A} : [0, T] \rightarrow \mathbb{R}^{2 \times 2}$, $\widehat{B} : [0, T] \rightarrow \mathbb{R}^{2 \times 1}$ and $\widehat{C} : [0, T] \rightarrow \mathbb{R}$ are the unique solution of the system of ordinary differential equations (A4)–(A5)–(A6) on $[0, T]$. Then, for an investor with CRRA utility preferences with risk aversion $\gamma < 1$, and risk penalization parameter $\varepsilon \geq 0$, the value function is of the form*

$$V(t, z, m, \widehat{\mu}) = \frac{z^\gamma}{\gamma} \exp \left(\frac{1}{2} (m, \widehat{\mu}) \widehat{A}(t) \begin{pmatrix} m \\ \widehat{\mu} \end{pmatrix} + \widehat{B}(t)^\top \begin{pmatrix} m \\ \widehat{\mu} \end{pmatrix} + \widehat{C}(t) \right).$$

Moreover, the optimal investment strategy is a stochastic process with Markovian feedback form $(\widehat{h}_t^*)_{t \in [0, T]} = (\widehat{h}^*(t, M_t, \widehat{\mu}_t))_{t \in [0, T]}$, where the function $\widehat{h}^*(t, m, \widehat{\mu})$ is given by

$$\widehat{h}^*(t, m, \widehat{\mu}) = \frac{\|\sigma_S\| \widehat{\Sigma}(t)^\top \widehat{B}(t) - r + (\xi + \frac{1}{2} (\widehat{A}(t) + \widehat{A}(t)^\top) \widehat{\Sigma}(t) \|\sigma_S\|) \begin{pmatrix} m \\ \widehat{\mu} \end{pmatrix}}{(1 - (\gamma - \varepsilon)) \|\sigma_S\|^2}.$$

Proof The proof is provided in Appendix 2. ■

The admissibility of the strategy $(\widehat{h}_t^*)_{t \in [0, T]} = (\widehat{h}^*(t, M_t, \widehat{\mu}_t))_{t \in [0, T]}$ follows from a similar argument as in Remark 3.3.

5. Discussion of results

In what follows, we analyze the solution of the optimal control problem. In particular, we will provide a comparison between strategies corresponding to full- and partial-information settings. Moreover, we will examine the value of information in the partial information setting.

5.1. Optimal trading strategies

The following corollary specifies the form of the optimal investment strategy under partial information.

COROLLARY 5.1 *Let $A_{ij} = A^{(ij)}(t)$ and $B_i = B^{(i)}(t)$ and $\widehat{A}_{ij} = \widehat{A}^{(ij)}(t)$ and $\widehat{B}_i = \widehat{B}^{(i)}(t)$. Optimal investment strategy under partial information is a stochastic process with Markovian*

feedback form $(\widehat{h}_t^)_{t \in [0, T]} = (\widehat{h}^*(t, M_t, \widehat{\mu}_t))_{t \in [0, T]}$, where the function $\widehat{h}^*(t, m, \widehat{\mu})$ has the form*

$$\widehat{h}^*(t, m, \widehat{\mu}) = \frac{h_m(m, \widehat{\mu}) + h_f(t, m, \widehat{\mu}) + h_p(t, m, \widehat{\mu})}{\|\sigma_S\|^2 (1 - (\gamma - \varepsilon))},$$

and

$$h_m(m, \widehat{\mu}) = \phi m + (1 - \phi) \widehat{\mu} - r,$$

$$h_f(t, m, \widehat{\mu})$$

$$= \|\sigma_S\|^2 \left(A_{11}(t)m + \frac{A_{12}(t) + A_{21}(t)}{2} \widehat{\mu} + B_1(t) \right) + \sigma_S^\top \sigma_\mu \left(A_{22}(t) \widehat{\mu} + \frac{A_{12}(t) + A_{21}(t)}{2} \widehat{\mu} + B_2(t) \right),$$

$$h_p(t, m, \widehat{\mu})$$

$$= \|\sigma_S\|^2 \left((\widehat{A}_{11}(t) - A_{11}(t))m + \frac{\widehat{A}_{12}(t) - A_{12}(t) + \widehat{A}_{21}(t) - A_{21}(t)}{2} \widehat{\mu} + (\widehat{B}_1(t) - B_1(t)) \right) + \sigma_S^\top \sigma_\mu \left((\widehat{A}_{22}(t) - A_{22}(t)) \widehat{\mu} + \frac{\widehat{A}_{12}(t) - A_{12}(t) + \widehat{A}_{21}(t) - A_{21}(t)}{2} \widehat{\mu} + (\widehat{B}_2(t) - B_2(t)) \right) + (1 - \phi)R(t) \left(\widehat{A}_{22}(t) \widehat{\mu} + \frac{\widehat{A}_{12}(t) + \widehat{A}_{21}(t)}{2} m + \widehat{B}_2(t) \right).$$

In the given context, h_m represents the myopic component of the optimal investment strategy. The term h_f corresponds to the portion of the optimal strategy that arises from intertemporal hedging demand, specifically addressing adverse changes in investment opportunities over time.

The component h_p captures the intertemporal hedging demand that emerges due to uncertainty in the available information. Our results show that this information-hedging component of the strategy changes linearly with the conditional error variance of the Kalman filter. This means that when the filter becomes less precise, due to factors such as higher volatility (noise variance) in the observed returns or a rise in the uncertainty about the latent state, the strategy adjusts by placing more weight on hedging against estimation errors. As a result, the total position deviates from a pure momentum-based allocation, reflecting the model's attempt to compensate for reduced confidence in the signal. In extreme cases, the overall position can turn negative even though prices are still going up. This sets the stage for a sudden loss when the market changes direction. Large filtering errors can

therefore lead to the quick sell-offs and sharp reversals seen during *momentum crashes*.

Finally, it is important to note that the effect of risk penalization manifests as an increase in the effective risk aversion parameter by an increment of ε , thereby modifying the portfolio manager's sensitivity to risk.

5.2. Information value

Several criteria in the literature have been used to address the value of information. For example Lee and Papanicolaou (2016), evaluates the loss of utility due to not being able to access all information regarding the underlying processes that govern their portfolio wealth. This quantity is defined as

$$L(t, z) = \mathbb{E} [V^{\text{full}}(t, Z_t, \mu_t, M_t) - V(t, Z_t, \hat{\mu}, M_t) | (Z_t = z) \vee \mathcal{F}_t].$$

Although the loss of utility provides insights on the comparison of full and partial information settings, it is not a monetary measure. Monetary measures have been proposed in Amendinger *et al.* (2003) or Chau *et al.* (2020). In this work, we use a similar concept as in Ceci and Colaneri (2025) and propose another alternative measure for a comparison, the so-called *indifference value*. This is defined as the monetary amount that an investor with partial information is willing to pay to implement the strategy under full information. As the set of feasible strategies under full information is strictly larger than the set of feasible strategies under partial information, the following inequality holds

$$\mathbb{E} [V^{\text{full}}(0, Z_0, \mu_0, M_0)] \geq \mathbb{E} [V(0, Z_0, \mu_0, M_0)] \quad (13)$$

Finding the indifference value amounts to solving the following equation

$$\mathbb{E} [V^{\text{full}}(0, Z_0, \mu_0, M_0)] = \mathbb{E} [V(0, Z_0 + \Delta Z_0, \mu_0, M_0)]$$

for ΔZ_0 . Notice that $\Delta Z_0 \geq 0$, because of (13) and the fact that the expected utility is increasing with respect to the initial wealth.

6. A one-factor model

In the following, we will investigate the case $\omega = \kappa$, which allows for interesting considerations. To this, we let

$$\Gamma_t := \phi M_t + (1 - \phi)\mu_t, \quad t \in [0, T], \quad (14)$$

that is, Γ represents the drift of price process in (3). Under the global filtration, Γ has the dynamics,

$$d\Gamma_t = (\phi(\phi - \omega)M_t + (1 - \phi)((\phi - \kappa)\mu_t + \kappa\bar{\mu})) dt + (\phi\sigma_S + (1 - \phi)\sigma_\mu)^\top dW_t.$$

Now we assume $\kappa = \omega$, then the market model reduces to a one-factor model driven by the factor process Γ with the

dynamics:

$$d\Gamma_t = \delta_t dt + \sigma_\Gamma^\top dW_t,$$

where $\delta_t = (1 - \phi)\omega\bar{\mu} - \Gamma_t(\omega - \phi)$, and $\sigma_\Gamma = \phi\sigma_S + (1 - \phi)\sigma_\mu$. Notice that the process Γ is Ornstein-Uhlenbeck, mean reverting around its target mean $\frac{1-\phi}{\omega-\phi}\omega\bar{\mu}$, with mean reversion speed $\omega - \phi$. That not only allows for simplified results from the mathematical point of view but also provides insights about our model. Setting $\omega = \kappa$ implicitly establishes a relationship between short-time and long-time effects, respectively accounted by the processes M and μ . From the dynamic of the momentum we see that this is mean-reverting around the factor Γ , scaled by a constant ω , and having speed of mean reversion given by ω as well,

$$dM_t = \omega \left(\frac{1}{\omega} \Gamma_t - M_t \right) dt + \sigma_S^\top dW_t.$$

The process μ_t is still mean reverting around the target mean $\bar{\mu}$, with the same speed of mean reversion of M . Moreover, we also observe that the drift of the stock price, in this case, is described in terms of a single mean-reverting process, i.e. Γ , as

$$dS_t = S_t \Gamma_t dt + S_t \sigma_S^\top dW_t. \quad (15)$$

In view of this and given that $\Gamma_t = \phi M_t + (1 - \phi)\mu_t$, we get that if $\omega = \kappa$ the momentum and long-run factor μ contribute to the drift of asset price with opposite effects, but with the same speed of mean reversion, and hence, in some sense, they *compensate*.

REMARK 6.1 (Interpretation of the cases $\omega < \kappa$ and $\omega > \kappa$) In this section the condition $\omega = \kappa$ implies that the short-run momentum component and the long-run mean-reversion component adjust at exactly the same speed. In that knife-edge case, the two forces compensate: momentum decays at the same rate as the underlying mean reversion pulls the drift back to fundamentals, leading to a single effective factor Γ .

When $\omega > \kappa$, (i.e. momentum decays faster than the mean-reversion factor adjusts), the momentum signal responds quickly to recent price movements while the long-run component reacts more gradually. Intuitively, the market exhibits short-lived bursts of trend-following on top of a slow-moving fundamental anchor. Momentum becomes a high-frequency predictor, while mean reversion operates on slower horizons. As a result, the drift can overshoot in the short run, e.g. due to strong recent returns, before gradually being pulled back by a more persistent, slowly evolving fundamental signal.

When $\omega < \kappa$ (i.e. momentum decays more slowly than mean reversion adjusts), the long-run factor reverts to its equilibrium more quickly than the momentum component fades. The fundamental mean-reversion signal thus becomes the fast force, rapidly correcting deviations from the long-run mean. Momentum, by contrast, becomes a sticky predictor, retaining memory of past returns for longer. The drift exhibits a stronger and more immediate stabilizing pull, with momentum behaving like a slow-moving trend that smooths over short-term fluctuations.

6.1. A one-factor model: the full information case

In the one-factor model, the wealth process satisfies the SDE

$$\frac{dX_t^{(h)}}{X_t^{(h)}} = (r + h_t(\Gamma_t - r)) dt + h_t \sigma_S^\top dW_t,$$

where $X_0 > 0$, and consequently, the penalized wealth has the dynamics

$$\frac{dZ_t^{(h)}}{Z_t^{(h)}} = \left(r + h_t(\Gamma_t - r) - \frac{1}{2} \varepsilon h_t^2 \|\sigma_S\|^2 \right) dt + h_t \sigma_S^\top dW_t.$$

This allows us to describe the optimal strategy and the optimal value function as follows.

PROPOSITION 6.2 Suppose that $\omega = \kappa$ and that $\gamma < \left(\frac{(\kappa - \phi) \|\sigma_S\|^2}{\|\sigma_S\|^2 + 2(\kappa - \phi) \sigma_S^\top \sigma_\Gamma} \right) / \left(1 + \frac{(\kappa - \phi) \|\sigma_S\|^2}{\|\sigma_S\|^2 + 2(\kappa - \phi) \sigma_S^\top \sigma_\Gamma} \right)$. Then, the optimal investment strategy under full information is a stochastic process with Markovian feedback form $(h_t)_{t \in [0, T]} = (h(t, \Gamma_t))_{t \in [0, T]}$, where the function $h(t, \Gamma)$ is given by

$$h(t, \Gamma) = \frac{\Gamma - r + (2A(t)\Gamma + B(t))\sigma_S^\top \sigma_\Gamma}{\|\sigma_S\|^2 (\varepsilon - (\gamma - 1))} \quad (16)$$

and the optimal value function is

$$V^{\text{full}}(t, z, \Gamma) = \frac{z^\gamma}{\gamma} v^{\text{full}}(t, \Gamma), \quad (17)$$

where $v^{\text{full}}(t, \Gamma) = \exp(A(t)\Gamma^2 + B(t)\Gamma + C(t))^\dagger$ and the functions $A, B, C : [0, T] \rightarrow \mathbb{R}$ are the unique solutions on any interval $[0, T]$ of the system of Riccati equations (A7)–(A8)–(A9) given in Appendix 3.

Proof The proof of this result is postponed to Appendix 3. ■

The admissibility of the optimal strategy follows from the same argument as in Remark 3.3. Note that in the current one-dimensional setting, it is possible to characterize the exact parameter conditions that ensure well-posedness of the corresponding Riccati system.

The optimal value function and the optimal strategies depend directly on the penalized wealth and on the factor Γ , which encompasses both the dependence on the momentum and the process μ . Notice that the value function is still homogeneous in the penalized wealth, and its structure is similar to the more general case.

6.2. A one-factor model: the partial information case

In the following, we will provide the results for the one-factor model under partial information.

First, we have the corollary providing the filter dynamics for the one-factor setting.

COROLLARY 6.3 Denote by $\hat{\Gamma}_t = \mathbb{E}[\Gamma_t | \mathcal{F}_t]$ the optional projection of Γ , defined in (14), with respect to filtration $\mathbb{F}$, a.s., for every $t \in [0, T]$ and by $R_t^\Gamma = \mathbb{E}[(\Gamma_t - \hat{\Gamma}_t)^2 | \mathcal{F}_t]$ the mean square error of estimation of filtering. We have:

$$\hat{\Gamma}_t = \phi M_t + (1 - \phi) \hat{\mu}_t,$$

$$R_t^\Gamma = (1 - \phi)^2 R_t,$$

for every $t \in [0, T]$.

Proof The result follows from Proposition 4.2, the properties of conditional expectation operator together with the linearity of Γ in μ and M , and that M is $\mathbb{F}$ adapted. ■

Now we are ready to provide the corresponding optimization result.

PROPOSITION 6.4 Suppose that $\omega = \kappa$. Then, the optimal investment strategy under partial information is a stochastic process with Markovian feedback form $(\hat{h}_t)_{t \in [0, T]} = (\hat{h}(t, \Gamma_t))_{t \in [0, T]}$, where the function $\hat{h}(t, \Gamma)$ is given by

$$\hat{h}(t, \hat{\Gamma}) = \frac{\hat{\Gamma} - r + (2\tilde{A}(t)\hat{\Gamma} + \tilde{B}(t)) \|\sigma_S\| \sigma_{\hat{\Gamma}, t}}{\|\sigma_S\|^2 (\varepsilon - (\gamma - 1))}, \quad (18)$$

where $\sigma_{\hat{\Gamma}, t} = \phi \|\sigma_S\| + \frac{(1 - \phi) \sigma_S^\top \sigma_\mu + R_t^\Gamma}{\|\sigma_S\|}$, and the optimal value function is given by

$$V(t, z, \hat{\Gamma}) = \frac{z^\gamma}{\gamma} v(t, \hat{\Gamma}), \quad (19)$$

where $v(t, \hat{\Gamma}) = \exp(\tilde{A}(t)\hat{\Gamma}^2 + \tilde{B}(t)\hat{\Gamma} + \tilde{C}(t))$ and the functions $\tilde{A}, \tilde{B}, \tilde{C} : [0, T] \rightarrow \mathbb{R}$ are the unique solutions of a system of Riccati equations given in Appendix 4.

Proof The proof of this result is provided in Appendix 4. ■

We can show the admissibility of the optimal strategy via the same lines of Remark 3.3.

6.3. Link between the optimal investment problem under full and partial information and value of information

In the one factor case, i.e. where $\omega = \kappa$, we can make use of the results given in Brendle (2006, Section 4)‡ to derive a relationship between value functions under full and partial information. In particular, we have that

$$\tilde{A}(t) = (1 - 2A(t)R_t^\Gamma)^{-1}A(t)$$

$$\tilde{B}(t) = (1 - 2A(t)R_t^\Gamma)^{-1}B(t)$$

$$\begin{aligned} \tilde{C}(t) = & C(t) + \frac{1}{2}(1 - 2R_t^\Gamma A(t))^{-1}R_t^\Gamma B^2(t) \\ & - \frac{1 - \gamma}{2} \log(1 - 2R_t^\Gamma A(t)) \end{aligned}$$

† With a slight abuse of notation for the one-factor model we still indicate the value function by V^{full} and the non-homogeneous component with v^{full} .

‡ To get the result we use the following identities in the notation of Brendle (2006), $d = 1$, $n = 2$, $\Sigma_s = \|\sigma_S\|^2$, $\Sigma_f = \|\sigma_\Gamma\|^2$, $R = \sigma_S^\top \sigma_\Gamma$ and $\lambda = \omega - \phi$.

$$-\frac{\gamma}{2} \log(1 - 2R_t^\Gamma L(t)) - \gamma F(t),$$

where $L(t)$ is the solution of the Riccati ordinary differential equation

$$\begin{aligned} \frac{dL(t)}{dt} = & -2 \left(\|\sigma_\Gamma\|^2 - \frac{(\sigma_S^\top \sigma_\Gamma)^2}{\|\sigma_S\|^2} \right) L^2(t) + 2(\omega - \phi)L(t) \\ & + 2 \frac{\sigma_S^\top \sigma_\Gamma}{\|\sigma_S\|^2} L(t) + \frac{1}{2} \frac{1}{\|\sigma_S\|^2}, \end{aligned}$$

with boundary condition $L(T) = 0$ and $F(t)$ is a real-valued function such that

$$\frac{dF(t)}{dt} = - \left(\|\sigma_\Gamma\|^2 - \frac{(\sigma_S^\top \sigma_\Gamma)^2}{\|\sigma_S\|^2} \right) (A(t) - L(t)), \quad (20)$$

with $F(T) = 0$, with $\tilde{A}$, $\tilde{B}$, $\tilde{C}$ satisfying equations (A10), (A11), (A12) respectively.

6.3.1. Information value within the one-factor model. In view of the explicit relation between the value functions under full and partial information in a one-factor model (see equations (17) and (19)) we get the following information value:

$$\left(1 + \frac{\Delta Z_0}{Z_0}\right)^\gamma = \frac{\mathbb{E} \left[e^{A(0)\Gamma_0^2 + B(0)\Gamma_0} \right]}{e^{\tilde{A}(0)\tilde{\Gamma}_0^2 + \tilde{B}(0)\tilde{\Gamma}_0}} e^{C(0) - \tilde{C}(0)}.$$

If we assume that Γ_0 has Gaussian distribution with mean $\hat{\Gamma}_0$ and variance $R^\Gamma(0)$ we get that

$$\begin{aligned} \left(1 + \frac{\Delta Z_0}{Z_0}\right)^\gamma &= \frac{e^{\frac{2A(0)\hat{\Gamma}_0^2 + B^2(0)R^\Gamma(0) + 2\hat{\Gamma}_0 B(0)}{2(1 - 2A(0)R^\Gamma(0))} - \log(1 - 2A(0)R^\Gamma(0))}}{e^{\tilde{A}(0)\hat{\Gamma}_0^2 + \tilde{B}(0)\hat{\Gamma}_0}} e^{C(0) - \tilde{C}(0)}. \end{aligned}$$

Using the results of the previous section, we have that

$$\begin{aligned} & \exp \left(\frac{2A(0)\hat{\Gamma}_0^2 + B^2(0)R^\Gamma(0) + 2\hat{\Gamma}_0 B(0)}{2(1 - 2A(0)R^\Gamma(0))} - \log(1 - 2A(0)R^\Gamma(0)) + C(0) \right) \\ &= \left(\frac{1 - 2R_0^\Gamma L(0)}{1 - 2R_0^\Gamma A(0)} \right)^{\gamma/2} \exp(\gamma F(0)) \\ & \quad \times \exp \left(\tilde{A}(0)\hat{\Gamma}_0^2 + \tilde{B}(0)\hat{\Gamma}_0 + \tilde{C}(0) \right), \end{aligned}$$

and hence

$$\Delta Z_0 = Z_0 \left(\sqrt{\frac{1 - 2R_0^\Gamma L(0)}{1 - 2R_0^\Gamma A(0)}} \exp(F(0)) - 1 \right).$$

Note, that a similar expression in the one-factor model has also been obtained in Brendle (2006). Moreover, if the value

of Γ_0 is observed at time $t = 0$, meaning that $\Gamma_0 = \hat{\Gamma}_0$ and $R_0^\Gamma = 0$, using the link between coefficients of the value functions under full and partial information for the one factor model, A , B , C and $\tilde{A}$, $\tilde{B}$, $\tilde{C}$ we obtain that

$$\left(1 + \frac{\Delta Z_0}{Z_0}\right)^\gamma = e^{C(0) - \tilde{C}(0)} = e^{\gamma F(0)},$$

that leads immediately to

$$\Delta Z_0 = (e^{F(0)} - 1)Z_0,$$

where F solves equation (20). Due to $\frac{dF(t)}{dt} \leq 0$ (see, Brendle 2006) and the terminal condition $F(T) = 0$, we have that $F(0) \geq 0$. Therefore, ΔZ_0 is non-negative. Qualitative properties of the value of information are discussed in Section 7.1.

7. Model estimation and simulation

We now describe how to estimate the model under partial information and show the importance of accounting for the unobservable nature of the stochastic process μ via a simulation study. We assume the *one-factor model* introduced in Section 6, where $\kappa = \omega$, which reduces computational costs due to the link between the full and partial information settings discussed in Section 6.3.†

7.1. Simulation study

In this section, we discuss optimal strategies and the value function under varying levels of information and different assumptions regarding knowledge of model parameters. The filter for the unobservable factor Γ , as given in (4.2), assumes that the model parameters are known. In practice, however, these parameters are typically unknown and must be estimated from observed data. Addressing these two challenges—parameter uncertainty and latent-state filtering—requires distinguishing between *parameter estimation* and *filtering*, which are conceptually distinct tasks.

To estimate the noise variance σ_S and the parameters of the latent state process under partial information, we employ the identification technique which was first introduced in Ahmed and Radaideh (1995). We choose this method because it demonstrates rapid and stable convergence and conveniently provides estimates for both σ_Γ and σ_S . Further details regarding the parameter identification technique we used are provided in Appendix 5.

As an alternative, one could consider the expectation-maximization (EM) algorithm for parameter estimation under partial information (see, e.g. Dembo and Zeitouni 1986, Florchinger 2002), which, however, suffers from slow convergence and typically requires separate methods to estimate the

† This assumption can be empirically supported by regressing the asset returns $\frac{dS_t}{S_t}$ of S&P 500 index data on the momentum factor M and an approximation of the mean-reversion factor μ , proxied by the dividend yield. The highest *adjusted* R^2 is obtained when ω is close to κ , suggesting that the one-factor specification offers a reasonable approximation. Results are available upon request.

Table 1. True and estimated parameters under partial information for the simulation study.

	ω	ϕ	$\bar{\mu}$	σ_Γ	σ_S
true parameters	2.65	0.15	1	$[-2, 0.6]$	$[0.7, -0.5]$
estimates	2.57	0.15	0.74	$[-1.89, 0.62]$	$[0.73, -0.47]$

Notes: The parameter ϕ is fixed according to the value in the numerical analysis of Koijen *et al.* (2009) and not estimated.

diffusion parameters, as it relies on an absolutely continuous change of measure.

The results of our numerical experiments are summarized below. First, we simulate the state process Γ and the observation process S , as defined in equations (14) and (15), respectively. The simulations use the parameter set specified in the first row of table 1 over a time horizon of $T = 1$ year, with time increments of $\Delta t = 1/252$, corresponding to daily observations. Note that, in this setting, the simulation of the underlying system can be carried out exactly, thanks to its Gaussian structure.

The simulated stock price series is used as the observation process for implementing the inference methodology described in Appendix 5. The resulting parameter estimates under partial information are reported in the second row of table 1. We note that these estimates are reasonably close to the true parameter values, indicating the effectiveness of the estimation procedure.

Figure 1 compares the true trajectory of the state variable (simulated Γ) with the filter obtained using known parameters and the filter obtained using estimated parameters. The plot demonstrates that both filters—based on known and estimated parameters—track the unobservable state process closely and yield similar performance. This is largely due to the small parameter estimation error (see table 1). It is important to note that any estimation technique is sensitive to the choice of initial parameter values. A qualitative analysis of this dependence can be found, for example, in Ahmed and Radaideh (1995).

Next, we compare the optimal trading strategy under full information, given by equation (16), with the corresponding strategies under partial information for both known and estimated parameter values, computed using equation (18). Figure 2 illustrates trajectories of the optimal strategies in each case. The strategy under partial information with estimated parameters closely follows that obtained using known parameters, indicating robustness of the filtering and estimation procedure.

Moreover, the strategies under partial information tend to be more conservative than those derived under full information. This behavior likely reflects the increased uncertainty in the partial information setting. However, this interpretation should be made with caution, as it may also depend on parameter values, correlation structures, and the investor's risk aversion.

We conclude this section with a sensitivity analysis of the indifference value of information, ΔZ_0 , that is, the amount a partially informed investor would be willing to pay to attain the same utility as an investor with full information. In this

analysis, we assume that both types of investors know the true model parameters and observe the true value of the factor process at time $t = 0$.

Figure 3 displays the indifference value in the one-factor model introduced in Section 6.3.1, plotted against varying levels of $\|\sigma_S\|$, under both high and low correlations between the signal and the observation, and for different values of ϕ .

The left panel of figure 3 illustrates the sensitivity of ΔZ_0 to changes in $\|\sigma_S\|$, for correlation value $\rho = 0.71$. Specifically, we keep $\sigma_S^{(2)}$ fixed and vary $\sigma_S^{(1)}$. An increase in noise variance $\sigma_S^{(1)}$, leads to a reduction in signal-to-noise ratio. Consequently, estimating the signal via filtering becomes more difficult - hence information more costly - resulting in a larger ΔZ_0 .

The right panel of figure 3 shows that the value of information increases with the parameter ϕ . This is because a larger ϕ leads to a higher long-run conditional variance, $\bar{R}$. Since, in this numerical example, the indifference value is computed assuming $R_0 = 0$, and the conditional variance R_t increases deterministically toward $\bar{R}$, a higher ϕ implies that the partially informed investor perceives greater future uncertainty. As a result, they would pay more to get better information on the factor process, as it helps to reduce uncertainty and better estimate the asset's drift.

8. Data application

We now test the results on the optimal strategy and the value function on real data. In particular, we compare two frameworks. In a first step we consider a *naïve* investor who believes that de-meaned dividend yield data is a good proxy for the mean reversion factor μ_t , and hence calibrate an Ornstein-Uhlenbeck process via maximum likelihood estimation. The naïve investor pretends to have full information and trades according to the strategy given in equation (16).

In a second step, we consider a more *sophisticated* investor who acknowledges that the data for the mean reversion factor μ_t is unavailable and applies the results from the partial information setup given in (18) to determine their trading strategy. This type of investor uses the methodology described in Section 7.1 to obtain the parameter estimates of the unobservable factor process μ_t under partial information. Both investors carry out their analysis under the assumption of the one-factor model of Section 6.

8.1. Data description

We use daily data from the S&P 500 index ranging from the 1st of January, 1991 to the 31st of December, 2014. We then provide the *out of sample* trading strategy using the estimated parameters obtained from this training data for an investment horizon of five years, ranging from January 2015 to the 4th of November, 2020. The data set consists of the daily returns as well as the de-meaned logarithmic dividend yield. Adjusted close price data for S&P500 are reported in figure 4. The figure displays the data and indicates a pattern where periods

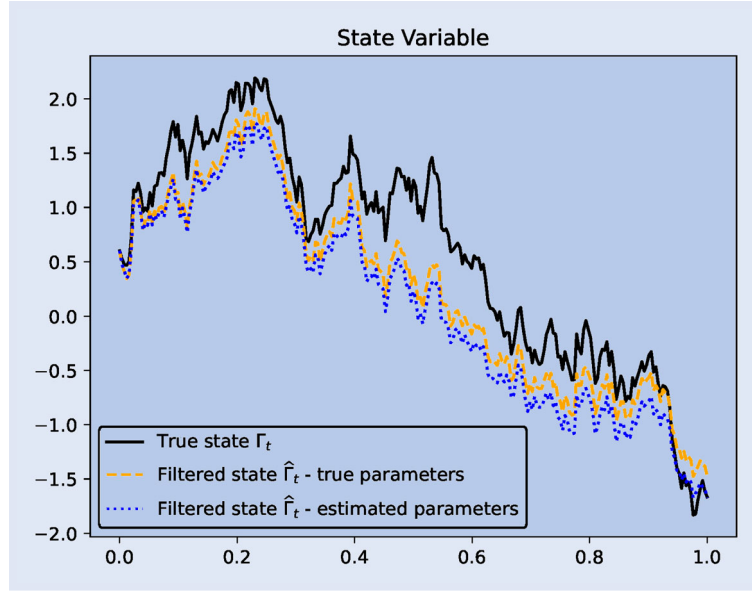


Figure 1. Comparison of the true state variable with the filtered state variables using both true and estimated parameters.

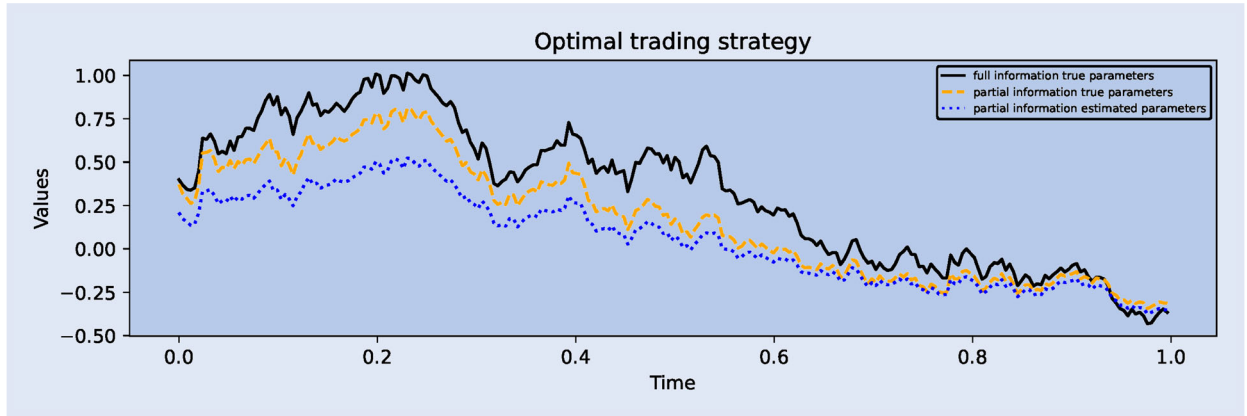


Figure 2. Comparison of the optimal trading strategies under full and partial information, both for true and estimated parameters in the simulation study.

of high returns tend to follow periods with high returns, likewise periods with low returns follow periods with low returns. This may suggest that momentum may be the dominant driver of the drift of S&P500 index. Consequently, the contribution of some unobservable process μ to the index drift seems to be relatively limited.

8.2. Parameter estimation

We let $\zeta = (\omega - \phi)$, $\eta = \frac{(1-\phi)\omega - \bar{\mu}}{(\omega - \phi)}$. The initial parameters for the estimation procedure $\alpha_0 = \{\zeta_0, \eta_0, \sigma_{\Gamma,0}^{(1)}, \sigma_{\Gamma,0}^{(2)}, \sigma_{s,0}^{(1)}, \sigma_{s,0}^{(2)}\}$ under partial information are obtained by calibrating the de-meaned logarithmic dividend yield to an Ornstein-Uhlenbeck

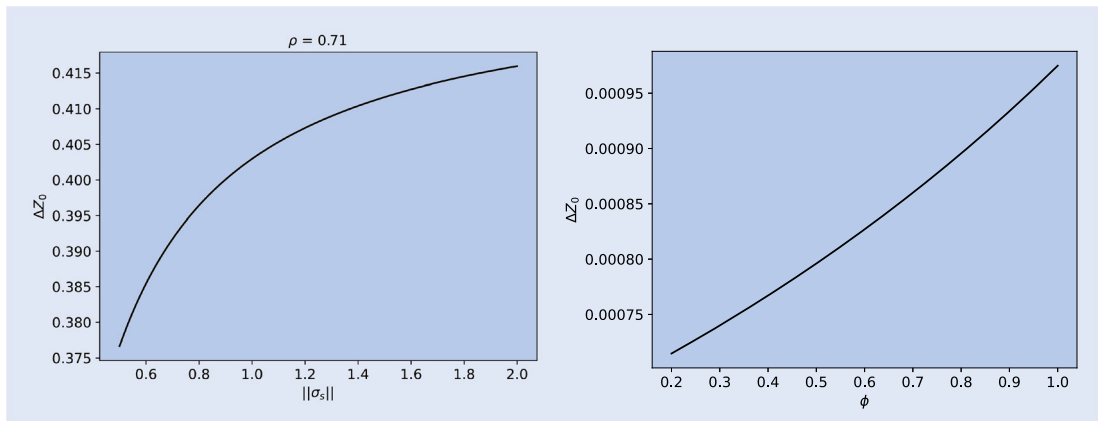


Figure 3. Value of information for different values of $\|\sigma_S\|$ and ϕ .

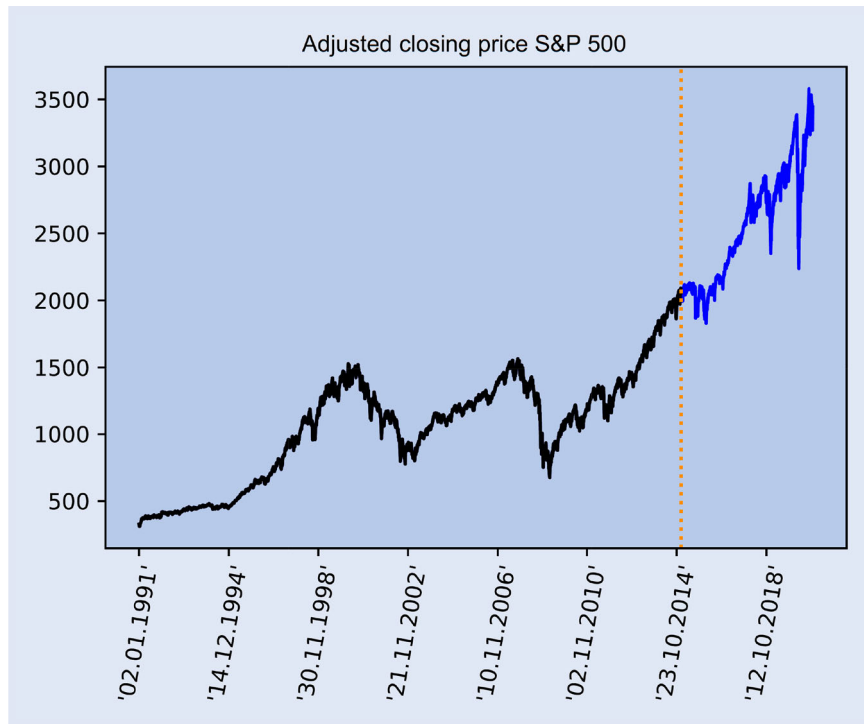


Figure 4. Visualization of adjusted close price of S&P500 index data. The dashed line indicates the beginning of the out-of-sample period.

Table 2. Estimated parameters for real data.

ζ	η	σ_{Γ}	σ_S
0.34	$-1.99\text{e}-03$	$[2.46\text{e}-06, 9.99\text{e}-03]$	$[0.16, -1.99\text{e}-03]$

process via maximum likelihood. A good initial temperature, as well as step size for the simulated annealing algorithm is chosen via an iteration over several possible starting values (see Appendix 5), and then we select the most beneficial initial temperature in terms of convergence and final loss value.

The estimated values for the parameters ζ , η , σ_{Γ} and σ_S are provided in table 2. Note that we do not need to identify ϕ , ω and $\bar{\mu}$ separately; however, as mentioned in Appendix 5, if an investor has expert knowledge on the value of ϕ , the parameters ω and $\bar{\mu}$ can be inferred from ζ and η .

In figure 5, we provide a comparison between the filtered mean reversion factor $\hat{\mu}$ and the de-measured dividend yield that is used by a naïve investor as a proxy for the mean reversion factor μ . We observe that the de-measured logarithmic dividend yield exhibits large volatility compared to the filtered estimate $\hat{\mu}$.

8.3. Investment strategies

In the following analysis, we present results on investment strategies for the naïve and the sophisticated investors for two levels of risk aversion, high ($\gamma = -3$) and low ($\gamma = -2$), setting $\varepsilon = 0$ and, for simplicity, assuming a zero risk-free rate ($r = 0$). Although the assumption of a zero risk-free rate is unrealistic, it does not affect the comparison between the two investor types, as changing its value would merely cause a

uniform shift in the strategy trajectories. The results, for an initial capital of one monetary unit, are shown in figure 6.

The trading strategy of the sophisticated investor, who accounts for the presence of an unobservable factor and decides according to rule given in equation (18), is smoother than that of the naïve investor, who approximates the latent factor with the de-measured dividend yield, pretending to have full information, and therefore acts according to equation (16). This difference arises because the filtered mean-reversion component, $\hat{\mu}$, is smoother than the trajectory of de-measured logarithmic dividend yields, used as approximations of the unobservable factor, by the two types of investors, respectively.

Moreover, the sophisticated investor tends to allocate a slightly larger share of wealth to the risky asset, and this behavior holds regardless of the level of risk aversion. As expected, lower risk aversion (higher γ) results in a more aggressive investment strategy, with a significantly higher allocation to the risky asset.

Figure 7 displays the corresponding value functions, which are monotonic decreasing as we get closer to maturity, as suggested by standard financial theory.

It is worth noting that these results may be influenced by market frictions such as short-selling constraints and transaction costs. However, abstracting from these considerations, the analysis suggests that a sophisticated investor would adopt a trading strategy that is locally significantly different from that of a naïve investor, although they share a similar general trend.

To better illustrate the difference in the values of the two investor types, it is useful to express the gap in their value functions in monetary terms. This can be done using the same methodology employed in the computation of the indifference value of information. Assuming an initial capital of one

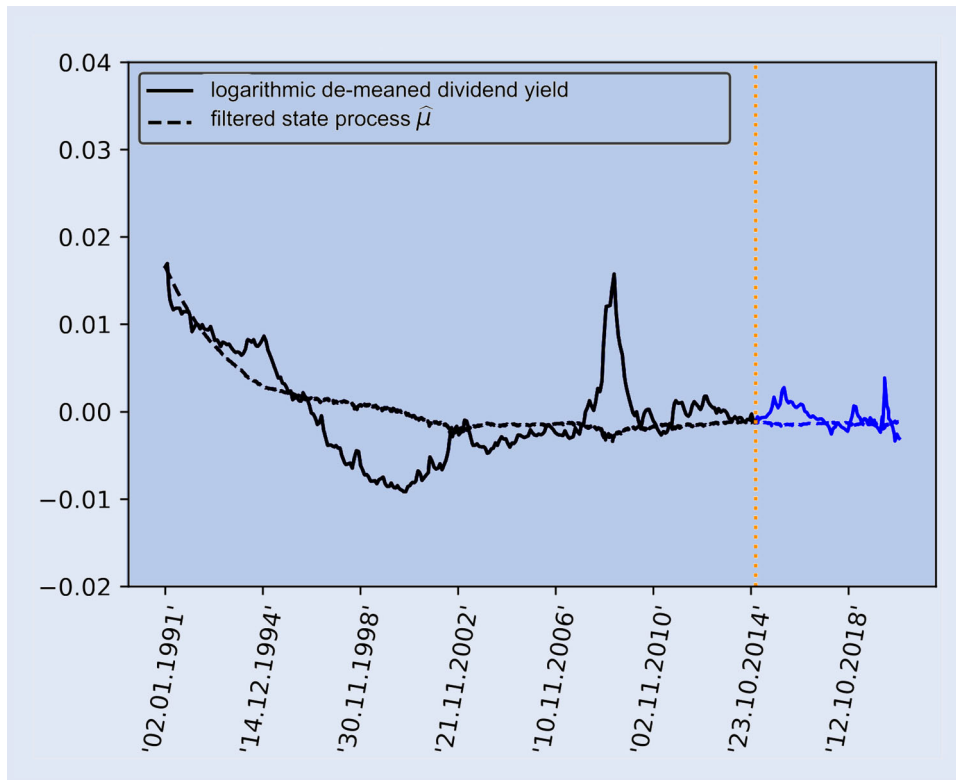


Figure 5. De-meaned logarithmic dividend yield (solid line) as a proxy for mean reversion factor and the filtered trajectory of the unobservable factor, $\hat{\mu}$ (dashed line).

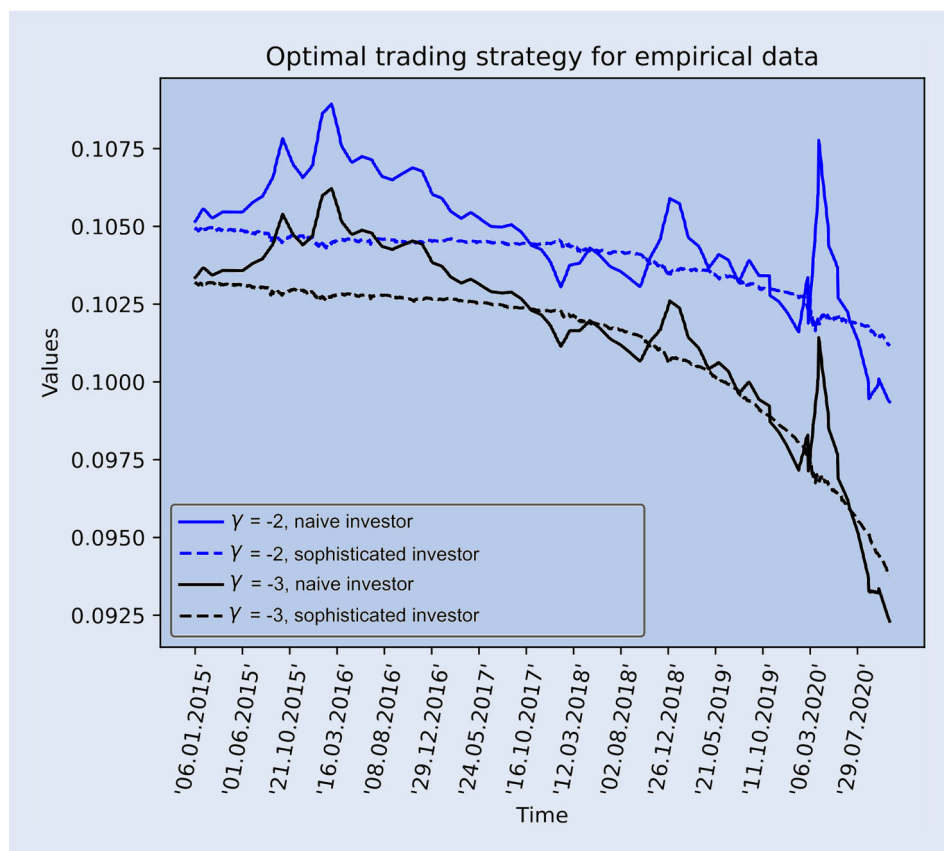


Figure 6. Comparison of the optimal trading strategies for the naïve and the sophisticated investors for the out-of-sample S&P 500 data under different risk aversion levels.

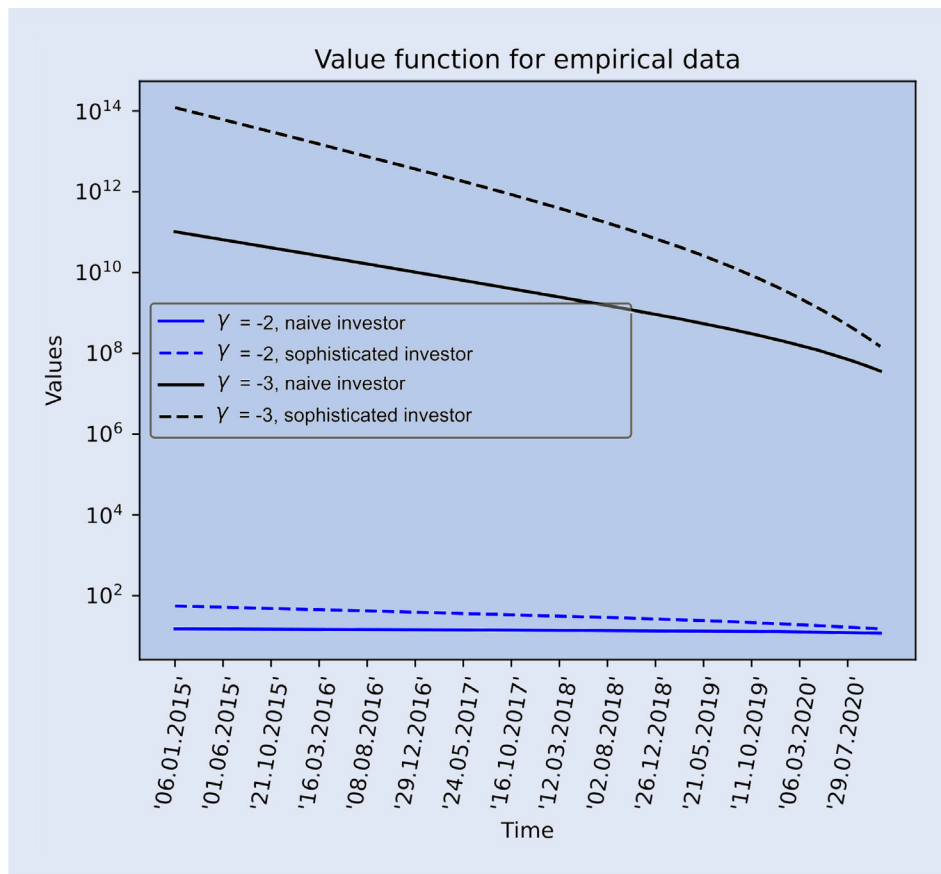


Figure 7. Comparison of the value function for the naïve and sophisticated investors for the out-of sample S&P 500 data under different levels of risk aversion.

monetary unit, the monetary indifference value is calculated to be -0.18 for the high risk aversion case and -0.13 for the low risk aversion case. It is crucial to stress that these numbers should not be interpreted as the value of information at time $t = 0$. Indeed, the naïve investor acts as if they observe the return of the S&P 500 index and applies decision rules based on full-information assumptions, but they do not really possess the information on the true latent factors. Therefore, the numbers reported above are meant to evaluate the difference in performance between the two investment strategies in monetary terms, rather than to provide the theoretical value of information.

9. Conclusion

To conclude, we studied a model in which returns of asset prices revolve around momentum and mean-reversion. These two fundamental factors are empirically observed and represent opposite trends in asset price dynamics, having an important impact on investment decisions. In particular, momentum has a dominant effect over shorter time horizons, while mean reversion drives returns over longer time horizons.

Unlike much of the existing literature, which approximates the unobservable mean-reversion component using observable fundamentals, such as the logarithmic dividend yield, we explicitly acknowledge its unobservability. We model the mean reversion factor as a latent process to be inferred from

asset price dynamics via filtering. Within this framework, we study a utility maximization problem under risk-penalized wealth for an investor endowed with partial information and characterize the corresponding optimal investment strategy.

Our theoretical analysis is supported by a simulation study and a real-data application to the S&P 500 index. The data application shows that an investor who accounts for partial information follows a smoother strategy than a naïve investor who assumes full information and approximates the mean-reversion factor using de-measured dividend yield data. Interestingly, the sophisticated investor achieves a higher value, highlighting the importance of accurately incorporating information constraints when making optimal investment decisions.

While our framework successfully integrates an observable momentum component with a latent mean-reverting factor under partial information, several limitations point toward fruitful extensions. A first natural step would be to extend the model to a multi-asset setting. Such an extension, however, is technically challenging: the state vector would grow dimensionally, the Kalman–Bucy filter would require tracking a full matrix Riccati equation, and the coupled HJB system would involve higher-dimensional quadratic forms, greatly increasing computational and analytical complexity. A second extension is to enrich the return dynamics by incorporating, e.g. stochastic volatility. This breaks the linear–Gaussian structure at the heart of our solution method, bringing, in particular to a setting in which the filter is no longer finite dimensional. Hence, the separation principle no longer yields

closed-form values and optimal controls. Such framework is also challenging from the estimation point of view, as it may require different techniques as, e.g. particle filters, that work in infinite dimensional settings. Finally, further extensions could consider alternative, empirically popular momentum definitions—such as fixed-window past returns, nonlinear technical indicators, or volatility-adjusted trend filters—although these are considerably harder to incorporate, as they typically break Markovianity or linear–Gaussian structure and therefore prevent the use of closed-form filtering and dynamic programming techniques. Despite these possible extensions, our analysis demonstrates that combining an observable momentum signal with a hidden mean-reverting component in a linear–Gaussian setting already captures key empirical features of return predictability while retaining full analytical tractability, offering a clear and rigorous benchmark for future developments.

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Appendices

Appendix 1. Proof of Theorem 3.2

We assume that the functions A , B and C are the unique solution of the system of ODEs

$$\begin{aligned} \frac{dA(t)}{dt} = & -2\gamma\theta_1\xi^\top + (1-\gamma+\varepsilon)\gamma\|\sigma_S\|^2\theta_1\theta_1^\top \\ & - \left(K^\top + \gamma\theta_1\|\sigma_S\|\Sigma^\top\right) \left(A(t) + A(t)^\top\right) \\ & - \frac{1}{4} \left(A(t) + A(t)^\top\right) \Sigma \Sigma^\top \left(A(t) + A(t)^\top\right), \end{aligned} \quad (A1)$$

$$\begin{aligned} \frac{dB(t)}{dt} = & -\gamma \left(\theta_0\xi^\top - r\theta_1^\top\right) + (1-\gamma+\varepsilon)\gamma\|\sigma_S\|^2\theta_0\theta_1^\top \\ & - B(t)K - \gamma B(t)^\top \Sigma \|\sigma_S\| \theta_1^\top \\ & - \frac{1}{2} \left(\kappa_0^\top + \gamma\theta_0\|\sigma_S\|\Sigma^\top\right) \left(A(t) + A(t)^\top\right) \\ & - \frac{1}{2} B(t)^\top \Sigma \Sigma^\top \left(A(t) + A(t)^\top\right), \end{aligned} \quad (A2)$$

$$\begin{aligned} \frac{dC(t)}{dt} = & -\gamma(1-\theta_0)r + \frac{1-\gamma+\varepsilon}{2}\gamma\theta_0^2\|\sigma_S\|^2 - \frac{1}{2}B(t)\Sigma\Sigma^\top \\ & - \left(\kappa_0^\top + \gamma\theta_0\|\sigma_S\|\Sigma^\top\right) B(t) \end{aligned}$$

$$- \frac{1}{4} \text{tr} \left(\left(A(t) + A(t)^\top \right) \Sigma \Sigma^\top \right), \quad (A3)$$

where $A(T) = 0_{2 \times 2}$, $B(T) = 0_{2 \times 1}$, $C(T) = 0$ and

$$\begin{aligned} \theta_0 &= \frac{\|\sigma_S\| \Sigma^\top B(t) - r}{(1-(\gamma-\varepsilon))\|\sigma_S\|^2}, \\ \theta_1 &= \frac{1}{(1-(\gamma-\varepsilon))\|\sigma_S\|^2} \left(\xi + \frac{1}{2} \left(A(t) + A(t)^\top \right) \Sigma \|\sigma_S\| \right). \end{aligned}$$

In what follows, we will use the following notation for partial derivatives: for every function $g : [0, T] \times \mathbb{R}_+ \times \mathbb{R} \rightarrow \mathbb{R}$, we write, for instance, $\frac{\partial g}{\partial t} = g_t$.

Assume that the value function of the optimization problem is continuously differentiable with respect to t and twice continuously differentiable with respect to (m, μ) . Then it solves the Hamilton-Jacobi-Bellman (HJB) equation given by

$$\begin{aligned} 0 = & V_t^{\text{full}}(t, z, m, \mu) \\ & + \sup_{h \in \mathcal{A}^G} \left\{ V_z^{\text{full}}(t, z, m, \mu) z \left(r + h(\phi m + (1-\phi)\mu - r) \right. \right. \\ & \left. \left. - \frac{\varepsilon}{2} h^2 \|\sigma_S\|^2 \right) \right. \\ & + V_\mu^{\text{full}}(t, z, m, \mu) \kappa(\bar{\mu} - \hat{\mu}) + V_m^{\text{full}}(t, z, m, \mu) ((\phi - \omega)m \\ & + (1-\phi)\mu) + \frac{1}{2} V_{zz}^{\text{full}}(t, z, m, \mu) z^2 h^2 \|\sigma_S\|^2 \\ & + \frac{1}{2} V_{\mu\mu}^{\text{full}}(t, z, m, \mu) \|\sigma_\mu\|^2 + \frac{1}{2} V_{mm}^{\text{full}}(t, z, m, \mu) \|\sigma_S\|^2 \\ & + V_{zm}^{\text{full}}(t, z, m, \mu) zh \|\sigma_S\|^2 \\ & \left. + V_{z\mu}^{\text{full}}(t, z, m, \mu) zh \sigma_S^\top \sigma_\mu + V_{\mu m}^{\text{full}}(t, z, m, \mu) \sigma_S^\top \sigma_\mu \right\} \end{aligned}$$

subject to the terminal condition $V(T, z, m, \mu) = \frac{z^\gamma}{\gamma}$, for all $(z, \mu, m) \in \mathbb{R}_+ \times \mathbb{R} \times \mathbb{R}$. We follow an argument based on a guess-and-verify approach. For the notation convenience we let $y^\top = (m, \mu)$ and define the ansatz

$$V^{\text{full}}(t, z, y) = \frac{z^\gamma}{\gamma} \exp \left(\frac{1}{2} y^\top A(t) y + B(t)^\top y + C(t) \right).$$

Maximizing over h yields the first-order condition and, after substituting the optimizer back into the HJB equation, the system (A1)–(A3). By the assumption of the theorem, this system admits a bounded solution (A, B, C) on $[0, T]$, so the candidate value function is $C^{1,2}$ and has at most quadratic growth in (m, μ) . Together with the moment estimates for (M, μ) and the integrability of h^* (see, Remark 3.3), this implies that under any admissible strategy, the local martingale terms in the Itô decomposition of $V^{\text{full}}(t, Z_t^{(h)}, Y_t)$ are true martingales on $[0, T]$. This is exactly the type of integrability condition used in the verification theorems for linear-Gaussian models; see, e.g. Davis and Lleo (2014).

Appendix 2. Proof of Theorem 4.4

We assume that the functions $\hat{A} \in \mathbb{R}^{2 \times 2}$, $\hat{B} \in \mathbb{R}^{2 \times 1}$ and $\hat{C} \in \mathbb{R}$ solve the following set of ordinary differential equations:

$$\begin{aligned} \frac{d\hat{A}(t)}{dt} = & -2\gamma\hat{\theta}_1\xi^\top + (1-\gamma+\varepsilon)\gamma\|\sigma_S\|^2\hat{\theta}_1\hat{\theta}_1^\top \\ & - \left(K^\top + \gamma\hat{\theta}_1\|\sigma_S\|\hat{\Sigma}^\top\right) \left(\hat{A}(t) + \hat{A}(t)^\top\right) \\ & - \frac{1}{4} \left(\hat{A}(t) + \hat{A}(t)^\top\right) \hat{\Sigma}(t) \hat{\Sigma}(t)^\top \left(\hat{A}(t) + \hat{A}(t)^\top\right), \end{aligned} \quad (A4)$$

$$\begin{aligned}
\frac{d\widehat{B}(t)}{dt} = & -\gamma \left(\widehat{\theta}_0 \xi^\top - r \widehat{\theta}_1^\top \right) + (1 - \gamma + \varepsilon) \gamma \|\sigma_S\|^2 \widehat{\theta}_0 \widehat{\theta}_1^\top \\
& - \widehat{B}(t) K - \frac{1}{2} \left(\kappa_0^\top + \gamma \widehat{\theta}_0 \|\sigma_S\| \widehat{\Sigma}(t)^\top \right) \left(\widehat{A}(t) + \widehat{A}(t)^\top \right) \\
& - \gamma \widehat{B}(t)^\top \widehat{\Sigma}(t) \|\sigma_S\| \widehat{\theta}_1^\top \\
& - \frac{1}{2} \widehat{B}(t)^\top \widehat{\Sigma}(t) \widehat{\Sigma}(t)^\top \left(\widehat{A}(t) + \widehat{A}(t)^\top \right), \tag{A5}
\end{aligned}$$

$$\begin{aligned}
\frac{d\widehat{C}(t)}{dt} = & -\gamma(1 - \widehat{\theta}_0)r + \frac{1 - \gamma + \varepsilon}{2} \gamma \widehat{\theta}_0^2 \|\sigma_S\|^2 \\
& - \frac{1}{2} \widehat{B}(t) \widehat{\Sigma}(t) \widehat{\Sigma}(t)^\top \\
& - \left(\kappa_0^\top + \gamma \widehat{\theta}_0 \|\sigma_S\| \widehat{\Sigma}(t)^\top \right) \widehat{B}(t) \\
& - \frac{1}{4} \text{tr} \left(\left(\widehat{A}(t) + \widehat{A}(t)^\top \right) \widehat{\Sigma}(t) \widehat{\Sigma}(t)^\top \right), \tag{A6}
\end{aligned}$$

where $\widehat{A}(T) = 0_{2 \times 2}$, $\widehat{B}(T) = 0_{2 \times 1}$, $\widehat{C}(T) = 0$ and

$$\widehat{\theta}_0 = \frac{\|\sigma_S\| \widehat{\Sigma}(t)^\top \widehat{B}(t) - r}{(1 - (\gamma - \varepsilon)) \|\sigma_S\|^2},$$

$$\widehat{\theta}_1 = \frac{1}{(1 - (\gamma - \varepsilon)) \|\sigma_S\|^2} \left(\xi + \frac{1}{2} \left(\widehat{A}(t) + \widehat{A}(t)^\top \right) \widehat{\Sigma}(t) \|\sigma_S\| \right).$$

Assume that the value function of the optimization problem is continuously differentiable with respect to its first argument and twice continuously differentiable with respect to the remaining arguments. Then, the value function solves the HJB equation given by

$$\begin{aligned}
0 = & V_t(t, z, m, \widehat{\mu}) \\
& + \sup_{h \in \mathcal{A}^F} \left\{ V_z(t, z, m, \widehat{\mu}) z \left(r + h(\phi m + (1 - \phi)\widehat{\mu} - r) \right. \right. \\
& \left. \left. - \frac{\varepsilon}{2} h^2 \|\sigma_S\|^2 \right) \right. \\
& + V_{\widehat{\mu}}(t, z, m, \widehat{\mu}) \kappa(\widehat{\mu} - \mu) + V_m(t, z, m, \widehat{\mu}) (1 - \phi)(\widehat{\mu} - M) \\
& + \frac{1}{2} V_{zz}(t, z, m, \widehat{\mu}) z^2 h^2 \|\sigma_S\|^2 \\
& + \frac{1}{2} V_{\widehat{\mu}\widehat{\mu}}(t, z, m, \widehat{\mu}) \frac{(\sigma_S^\top \sigma_\mu + (1 - \phi)R)^2}{\|\sigma_S\|^2} \\
& + \frac{1}{2} V_{mm}(t, z, m, \widehat{\mu}) \|\sigma_S\|^2 + V_{zm}(t, z, m, \widehat{\mu}) zh \|\sigma_S\|^2 \\
& + V_{z\widehat{\mu}}(t, z, m, \widehat{\mu}) zh(\sigma_S^\top \sigma_\mu + (1 - \phi)R) \\
& \left. + V_{\widehat{\mu}m}(t, z, m, \widehat{\mu})(\sigma_S^\top \sigma_\mu + (1 - \phi)R) \right\}
\end{aligned}$$

subject to the terminal condition $V(T, z, m, \widehat{\mu}) = \frac{z^\gamma}{\gamma}$, for all $(z, m, \widehat{\mu}) \in \mathbb{R}_+ \times \mathbb{R} \times \mathbb{R}$. We denote by $\widehat{y} = (m, \widehat{\mu})^\top$ the state variables and consider the ansatz,

$$V(t, z, \widehat{y}) = \frac{z^\gamma}{\gamma} \exp \left(\frac{1}{2} \widehat{y}^\top \widehat{A}(t) \widehat{y} + \widehat{B}(t)^\top \widehat{y} + \widehat{C}(t) \right).$$

Computing the first and second order conditions with respect to control, and, after substituting the optimizer back into the HJB equation, the system (B1)–(B3) follows.

By the assumption of Theorem 4.4, (B1)–(B3) admit a unique solution on $[0, T]$, which is continuous and thus bounded. Hence $V \in C^{1,2}$ with at most exponential–quadratic growth in $(m, \widehat{\mu})$. The conditional variance $R(t)$ is bounded and non-negative by Remark 4.3, and $(M_t, \widehat{\mu}_t)$ have bounded second moments. The same square-integrability argument as in Appendix A therefore shows that, for any $h \in \mathcal{A}^F$, the local martingale term arising in Itô's formula for $V(t, Z_t^{(h)}, M_t, \widehat{\mu}_t)$ is a true martingale. The admissibility (in particular, square-integrability) of the feedback control $\widehat{h}^*$ follows from an argument identical to Remark 3.3, now applied to the pair $(M_t, \widehat{\mu}_t)$.

The function V is $C^{1,2}$, satisfies the terminal condition, solves the HJB equation, and the supremum in the Hamiltonian is attained at the feedback control $\widehat{h}^*$. The martingale condition required by the verification theorem is ensured by the boundedness of $\widehat{A}, \widehat{B}, \widehat{C}$, the bounded second moments of $(M_t, \widehat{\mu}_t)$, and the integrability condition in Definition 2. Therefore, the verification result follows as in Appendix 1.

Appendix 3. Proof of Proposition 6.2

Following the same approach as in the proof of Theorem 3.2 (see Appendix 1) we get the value function and the optimal strategy for the one factor model given in (17) and (16), respectively, where the functions A, B and C are the unique solutions of the system

$$A_t = -\frac{\gamma (2A\sigma_S^\top \sigma_\Gamma + 1)^2}{2 \|\sigma_S\|^2 (\varepsilon - (\gamma - 1))} + 2A(\omega - \phi) - 2A^2 \|\sigma_\Gamma\|^2 \tag{A7}$$

$$\begin{aligned}
B_t = & -\frac{\gamma (2A\sigma_S^\top \sigma_\Gamma + 1) (B\sigma_S^\top \sigma_\Gamma - r)}{\|\sigma_S\|^2 (\varepsilon - (\gamma - 1))} - 2A(1 - \phi)\omega\bar{\mu} \\
& + B(\omega - \phi) - 2AB \|\sigma_\Gamma\|^2 \tag{A8}
\end{aligned}$$

$$\begin{aligned}
C_t = & -\gamma r - \frac{\gamma (B\sigma_S^\top \sigma_\Gamma - r)^2}{2 \|\sigma_S\|^2 (\varepsilon - (\gamma - 1))} - B(1 - \phi)\omega\bar{\mu} \\
& - \frac{1}{2} (2A + B^2) \|\sigma_\Gamma\|^2, \tag{A9}
\end{aligned}$$

with the final condition $A(T) = 0, B(T) = 0, C(T) = 0$. In the one-factor model we are in the same setting as in Brendle (2006), in which it is required that

$$\begin{aligned}
\gamma < & \left(\frac{(\kappa - \phi) \|\sigma_S\|^2}{\|\sigma_S\|^2 + 2(\kappa - \phi) \sigma_S^\top \sigma_\Gamma} \right) / \\
& \left(1 + \frac{(\kappa - \phi) \|\sigma_S\|^2}{\|\sigma_S\|^2 + 2(\kappa - \phi) \sigma_S^\top \sigma_\Gamma} \right),
\end{aligned}$$

such that the solutions of (A7), (A8), and (A9) exist. For $\gamma < 0$, existence and uniqueness of the solution of the system of equations (A7)–(A8)–(A9) on any interval $[0, T]$ can be shown through the following steps. Existence and uniqueness of the solution of (A7) follows from Filipović (2009, Lemma 10.2). Given this, existence and uniqueness of the solution of the linear ODE (A8) follows from Lipschitz continuity of function $A(t)$ (see, e.g. Teschl 2012, Section 2.2). Finally, iterating this argument, by Lipschitz continuity of $A(t)$ and $B(t)$, and we obtain existence and uniqueness of the solution of (A9). The verification result follows from the same arguments as in Appendix 1.

Appendix 4. Proof of Proposition 6.4

In this section we characterize the value function under partial information in the one factor model corresponding to the choice $\kappa = \omega$.

First, we write the dynamics of the underlying state process $(Z, \widehat{\Gamma})$. Corollary 6.3 together with (10) and (8) yield that

$$d\widehat{\Gamma}_t = ((\phi - \omega)\widehat{\Gamma}_t + (1 - \phi)\omega\bar{\mu}) dt + \sigma_{\widehat{\Gamma},t} dI_t,$$

where $\sigma_{\widehat{\Gamma},t} = \phi \|\sigma_S\| + \frac{(1 - \phi)\sigma_S^\top \sigma_\mu + R_t^\Gamma}{\|\sigma_S\|}$. The dynamics of Z is given by (see (11)):

$$\frac{dZ_t}{Z_t} = \left(r + h_t (\widehat{\Gamma}_t - r) - \frac{\varepsilon}{2} h_t^2 \|\sigma_S\|^2 \right) dt + h_t \|\sigma_S\| dI_t.$$

Here we again follow a guess-and-verify approach, and derive first the HJB equation for the value function. Suppose that the function V

is smooth, then it solves the HJB equation

$$0 = V_t(t, z, \hat{\Gamma}) + \sup_{h \in \mathcal{A}^x} \left\{ V_z(t, z, \hat{\Gamma}) z \left(r + h(\hat{\Gamma} - r) - \frac{\varepsilon}{2} h^2 \|\sigma_S\|^2 \right) \right. \\ \left. + V_{\hat{\Gamma}}(t, z, \hat{\Gamma}) ((\phi - \omega)\hat{\Gamma} + (1 - \phi)\omega\bar{\mu}) \right. \\ \left. + \frac{1}{2} V_{zz}(t, z, \hat{\Gamma}) z^2 h^2 \|\sigma_S\|^2 \right. \\ \left. + \frac{1}{2} V_{\hat{\Gamma}\hat{\Gamma}}(t, z, \hat{\Gamma}) \sigma_{\hat{\Gamma},t}^2 + V_{z\hat{\Gamma}}(t, z, \hat{\Gamma}) zh \|\sigma_S\| \sigma_{\hat{\Gamma},t} \right\}$$

We guess that $V(t, z, \hat{\Gamma}) = \frac{\gamma}{\gamma} v(t, \hat{\Gamma})$ where $v(t, \hat{\Gamma}) = \exp(\tilde{A}(t)\hat{\Gamma}^2 + \tilde{B}(t)\hat{\Gamma} + \tilde{C}(t))$. Then by substituting we obtain that the optimal strategy is given by

$$h(t, \hat{\Gamma}) = \frac{\Gamma - r + (2\tilde{A}(t)\hat{\Gamma} + \tilde{B}(t)) \|\sigma_S\| \sigma_{\hat{\Gamma},t}}{\|\sigma_S\|^2 (\varepsilon - (\gamma - 1))}$$

and that the functions $\tilde{A}, \tilde{B}$ and $\tilde{C}$ satisfy:

$$\tilde{A}_t = -\frac{\gamma \left(2\tilde{A}(t) \|\sigma_S\| \sigma_{\hat{\Gamma},t} + 1 \right)^2}{2 \|\sigma_S\|^2 (\varepsilon - (\gamma - 1))} + 2\tilde{A}(t)(\omega - \phi) - 2\tilde{A}(t)^2 \sigma_{\hat{\Gamma},t}^2 \quad (\text{A10})$$

$$\tilde{B}_t = -\frac{\gamma \left(2\tilde{A}(t) \|\sigma_S\| \sigma_{\hat{\Gamma},t} + 1 \right) (\tilde{B}(t) \|\sigma_S\| \sigma_{\hat{\Gamma},t} - r)}{\|\sigma_S\|^2 (\varepsilon - (\gamma - 1))} \\ - 2\tilde{A}(t)(1 - \phi)\omega\bar{\mu} + \tilde{B}(t)(\omega - \phi) - 2\tilde{A}(t)\tilde{B}(t)\sigma_{\hat{\Gamma},t}^2 \quad (\text{A11})$$

$$\tilde{C}_t = -\gamma r - \frac{\gamma (\tilde{B}(t) \|\sigma_S\| \sigma_{\hat{\Gamma},t} - r)^2}{2 \|\sigma_S\|^2 (\varepsilon - (\gamma - 1))} - \tilde{B}(t)(1 - \phi)\omega\bar{\mu} \\ - \frac{1}{2} (2\tilde{A}(t) + \tilde{B}(t)^2) \sigma_{\hat{\Gamma},t}^2 \quad (\text{A12})$$

The verification result follows from the same arguments as in Appendix 2.

Appendix 5. Details on estimation methodology

In this section, we provide the details of the estimation methodology, which relies on the results given in Ahmed and Radaideh (1995).

We start by discretizing the system consisting of (15), (14). Consider a finite time interval $[0, T]$ that is split into m subintervals such that $\Delta t = \frac{T}{m}$ and put $\Delta Z_t = W_{t+\Delta t} - W_t$. Then $\Delta Z_t \sim \mathcal{N}(0, I_2 \sqrt{\Delta t})$, where I_2 denotes the 2×2 identity matrix, and we consider the following discrete time approximation to the above system

$$S_{t+\Delta t} = S_t \exp \left(\left(\Gamma_t - \frac{\|\sigma_S\|^2}{2} \right) \Delta t + \sigma_S^\top \Delta Z_t \right) \\ \Gamma_{t+\Delta t} = \Gamma_t \exp(-\zeta \Delta t) + \eta(1 - \exp(-\zeta \Delta t)) \\ + \sqrt{\frac{1 - \exp(-2\zeta \Delta t)}{2\zeta}} \sigma_\Gamma^\top \Delta Z_t, \quad \Gamma_0 \sim \mathcal{N}(\eta, \bar{R}_\Gamma). \quad (\text{A13})$$

where $\zeta = (\omega - \phi)$, $\eta = \frac{(1-\phi)\omega\bar{\mu}}{(\omega-\phi)}$ and $\sigma_S \in \mathbb{R}^2$.

The problem we face is to identify the system parameters $\alpha = \{\zeta, \eta, \sigma_\Gamma^{(1)}, \sigma_\Gamma^{(2)}, \sigma_S^{(1)}, \sigma_S^{(2)}\}$ from the observation history $(S_t)_{0 \leq t \leq T}$. Note, that we do not need to identify ϕ , ω and $\bar{\mu}$ standing on their own, as they only appear in the form of ζ and η in $\tilde{A}_t$, $\tilde{B}_t$ and $\tilde{C}_t$. If an investor has expert knowledge on the value of ϕ , the parameters ω and $\bar{\mu}$ can be inferred from ζ and η . Let $\mathcal{P}$ denote the set of admissible parameters. Let α^* denote the true parameter vector

corresponding to system (A13) and let $\alpha \in \mathcal{P}$ denote any arbitrary admissible parameter vector. Recall that the conditional expectation of Γ , given $\mathcal{F}_t$, $\hat{\Gamma}_t = E[\Gamma_t | \mathcal{F}_t]$ is given by

$$d\hat{\Gamma}_t(\alpha) = ((1 - \phi)\omega\bar{\mu} - \hat{\Gamma}_t(\alpha)(\omega - \phi)) dt \\ + \frac{R(\alpha)(t) + \sigma_S^\top \sigma_\Gamma}{\|\sigma_S\|^2} \left(\frac{dS_t}{S_t} - \hat{\Gamma}_t(\alpha) dt \right), \\ \frac{dR^\Gamma(\alpha)(t)}{dt} = \|\sigma_\Gamma\|^2 - 2(\omega - \phi)R^\Gamma(\alpha)(t) - \frac{(R^\Gamma(\alpha)(t) + \sigma_S^\top \sigma_\Gamma)}{\|\sigma_S\|^2},$$

where we make the dependence on the chosen α explicit, using the above notation.

Since Γ is given as an Ornstein-Uhlenbeck process, we can write down the dynamics of expectation and variance as follows

$$d\bar{\Gamma}(\alpha)_t/dt = ((1 - \phi)\omega\bar{\mu} - \bar{\Gamma}_t(\alpha)(\omega - \phi)) dt, \quad \bar{\Gamma}(\alpha)_0 = \Gamma_0 \\ d\text{Var}[\Gamma(\alpha)(t)]/dt \\ = -2(\omega - \phi)\text{Var}[\Gamma(\alpha)(t)] + \|\sigma_\Gamma\|^2, \quad \text{Var}[\Gamma(\alpha)(0)] = R_0.$$

We have that by definition

$$R^\Gamma(\alpha)(t) = E[(\Gamma(\alpha)(t) - \hat{\Gamma}(\alpha)(t))^2] \\ = E[(\Gamma(\alpha)(t) - \bar{\Gamma}(\alpha)(t) + \bar{\Gamma}(\alpha)(t) - \hat{\Gamma}(\alpha)(t))^2] \\ = E[(\Gamma(\alpha)(t) - \bar{\Gamma}(\alpha)(t))^2] + E[(\bar{\Gamma}(\alpha)(t) - \hat{\Gamma}(\alpha)(t))^2] \\ + 2E[(\Gamma(\alpha)(t) - \bar{\Gamma}(\alpha)(t))(\bar{\Gamma}(\alpha)(t) - \hat{\Gamma}(\alpha)(t))] \\ = \text{Var}(\Gamma(\alpha)(t)) - E[(\hat{\Gamma}(\alpha)(t) - \bar{\Gamma}(\alpha)(t))^2], \quad (\text{A14})$$

where in the last equality we have used the measurability of $(\bar{\Gamma}(\alpha)(t) - \hat{\Gamma}(\alpha)(t))$ with respect to the history of S_t . Define $e(\alpha)(t) := \hat{\Gamma}(\alpha)(t) - \bar{\Gamma}(\alpha)(t)$. It is easy to see that $e(\alpha)$ satisfies the SDE

$$de_t(\alpha) = ((1 - \phi)\omega\bar{\mu} - e_t(\alpha)(\omega - \phi)) dt \\ + \frac{R^\Gamma(\alpha)(t) + \sigma_S^\top \sigma_\Gamma}{\|\sigma_S\|^2} \left(\frac{dS_t}{S_t} - \hat{\Gamma}_t(\alpha) dt \right).$$

Therefore we can write (A14) as

$$R^\Gamma(\alpha)(t) = \text{Var}(\Gamma(\alpha)(t)) - E[e(\alpha)(t)^2].$$

Suppose that we have observations $(S_t)_{0 \leq t \leq T}$, which are generated from the true system parameters α^* . If we now take these observations as an input in the innovations process appearing in the dynamics of $\hat{\Gamma}(t)$ and thus also in (A14), with an arbitrary choice of α it is clear that (A14) does not hold. It is only true if the trial parameter vector α is chosen to be the true parameter vector α^* . Therefore the natural idea is to seek a $\alpha^* \in \mathcal{P}$ that minimizes the cost functional J given by

$$J(T, \alpha) := \int_0^T (R^\Gamma(\alpha)(t) + R^e(\alpha)(t) - \text{Var}(\Gamma(\alpha)(t)))^2 dt, \quad (\text{A15})$$

where $R^e(\alpha)(t) := E[(\hat{\Gamma}(\alpha)(t) - \bar{\Gamma}(\alpha)(t))^2]$. Note that $\hat{\Gamma}(\alpha)(t)$ is the only quantity that depends on the observations S in this cost functional. Then, one aims to minimize $J(T, \alpha)$ by finding the optimal set of parameters α^* .

If we approximate $R^e(\alpha)$ by, e.g. the classical Monte Carlo estimator, we have an approximate solution for the parameter inference problem. Figure A1 provides an intuition on how the number of simulated Monte Carlo paths used to approximate $R^e(\alpha)$ influences the deviation from the true $R(\alpha)(t)$, even if we use the ground truth parameters.

The discretized version of cost function (A15) can be minimized using any numerical scheme, suited for minimization of multivariate objectives. We choose simulated annealing, which is a gradient-free

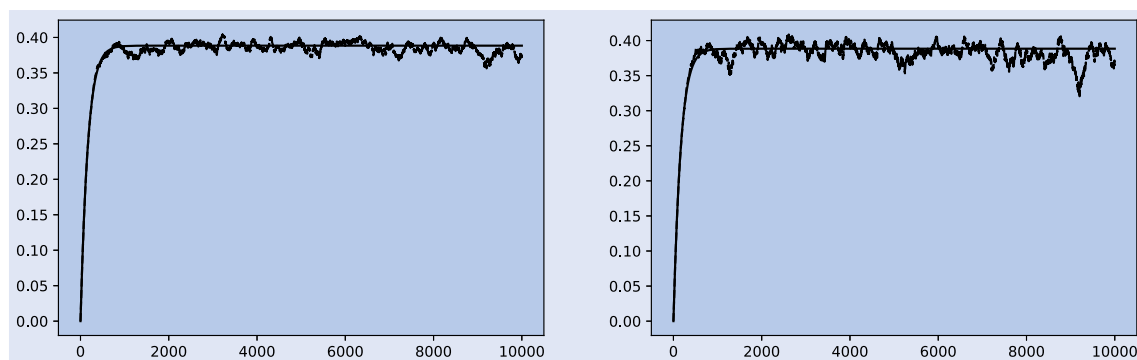


Figure A1. Illustration of variation of Monte Carlo error in A15, using different number of simulation paths (Left panel: 120 simulation paths; right panel 50 simulation paths).

method, as this algorithm has been proposed in the optimization of (A15) in Ahmed and Radaideh (1995). The difficulty here is, of course, that in the real-data situation, one only has access to one

historical trajectory of S . Therefore, when working with real data the cross-sectional expectation $R^e(\alpha)(t)$ has to be approximated using time averages.